

Calibres of Compact Spaces

In this chapter we study the calibres of compact spaces (and the precalibres of all spaces); the surprising fact is that an extensive class of cardinals, depending only on the Souslin number of the compact space in question, is a class of calibres for the space. This is the content of Theorems 5.6 and 5.25, the two main results of this section. (When the generalized continuum hypothesis is assumed, these results assume a particularly simple form, given in Corollary 5.26.) The techniques used for the proof of these results are combinatorial, and are contained in Lemmas 5.1 (a ramification method for regular cardinals), 5.4 (a method that deals with singular cardinals), and 5.22 (dealing with the ‘exceptional’ cardinals). It is noted that the combinatorial techniques used here involve ramification methods more general and more powerful than those given by the Erdős–Rado theorem on quasi-disjoint families of sets and its extension to singular cardinals; as to the determination of cardinal invariants, these latter results are useful essentially only for product spaces or for spaces with a product-like structure, while we in the present section are able to examine spaces quite general in nature. On the basis of examples given in Theorems 5.28 and 6.18, we can claim that the results are essentially all that can be proved for general spaces (see the diagrams in 7.19).

For purposes of motivation we note that the results of this section are closely related to a number of results, also due to Argyros and Tsarpalias and stated in some detail in the Notes, deriving from the following question concerning Banach spaces: When can the Banach space l_α^1 of absolutely summable real functions defined on (a set of cardinality) α be embedded isomorphically into every closed, linear α -dimensional subspace of the Banach space $C(X)$ of real-valued, continuous functions on the compact space X ? The appropriate condition, roughly speaking, is that α and the Souslin number κ of X

are in the cardinal relation given in 5.6 (and analogously for 5.25). While these functional analytic proofs are more intricate, they also rely on the same combinatorial methods, together with the reduction (given by Rosenthal) of the embeddability of l_α^1 to a combinatorial question on the existence of certain (Boolean) independent families of pairs of open subsets of X ; this question in turn is related to (but more delicate than) the question of the existence of a family of open subsets of X with non-empty intersection appearing in the concept of calibre for X .

Byproducts of Theorem 5.6 are: (a) the impossibility of finding, for a given class $\mathbf{T}$ of cardinals satisfying the natural conditions given in the statement of Corollary 4.7, a compact space whose class of non-calibres is exactly $\mathbf{T}$; (b) assuming the generalized continuum hypothesis, the reduction of the question of determining all calibres of a compact space to the consideration of regular cardinals only (Corollary 5.9); and (c) again assuming the generalized continuum hypothesis, a satisfactory product theorem for singular (as well as regular) cardinals (Corollary 5.10).

These results have analogues concerning calibres of compact spaces X with λ^+ -complete topologies for $\lambda < S(X)$ (Theorem 5.16) and concerning calibres of λ^+ -box topologies on product spaces $\prod_{i \in I} X_i$ with λ a cardinal less than a regular upper bound for the numbers $S(X_i)$, $i \in I$ (Theorem 5.21); the regular cases of these theorems strengthen results given earlier by Juhász (Corollary 5.13) and by Kurepa and Hedrlín (Corollary 5.18), respectively.

Strongly $S(X)$ -inaccessible cardinals as calibres

Definition. Let X be a space and $\{U_\xi : \xi < \alpha\}$ a set of non-empty, open subsets of X . An index $\xi < \alpha$ is a *representative* of $\{U_\xi : \xi < \alpha\}$ if the following condition is satisfied:

If $\beta < \alpha$ and $\{W_\zeta : \zeta < \beta\}$ is a set of non-empty, open subsets of U_ξ , then there is $A \subset \alpha$ with $|A| = \alpha$ such that $\{U_\xi \cap W_\zeta : \xi \in A\}$ has the finite intersection property for all $\zeta < \beta$.

5.1 Lemma. Let $\omega \leq \kappa \ll \alpha$ with α regular and let X be a space such

that $S(X) \leq \kappa$. Then every set $\{U_\xi : \xi < \alpha\}$ of non-empty, open subsets of X has a representative.

Proof. Suppose that $\{U_\xi : \xi < \alpha\}$ is a set of non-empty, open subsets of X with no representative. For $A \subset \alpha$ with $|A| = \alpha$ we choose $\xi_A \in A$. There are

a cardinal $\beta_A < \alpha$, and

a set $\{W_\zeta^A : \zeta < \beta_A\}$ of non-empty, open subsets of U_{ξ_A} , such that

if $B \subset A$ and $\{U_\xi \cap W_\zeta^A : \xi \in B\}$ has the finite intersection property for all $\zeta < \beta_A$, then $|B| < \alpha$.

For $A \subset \alpha$ with $|A| = \alpha$ we set

$\mathcal{B}_A = \{B \subset A : \{U_\xi \cap W_\zeta^A : \xi \in B\} \text{ has the finite intersection property for all } \zeta < \beta_A\}$.

Since the set $\mathcal{B}_A$ partially ordered by inclusion is inductive, there is a maximal element B_A of $\mathcal{B}_A$. We have $|B_A| < \alpha$, and since $\{\xi_A\} \in \mathcal{B}_A$ we have $B_A \neq \emptyset$.

For $\xi \in A \setminus B_A$ it follows from the maximality of B_A that there are a finite, non-empty subset $F(\xi)$ of B_A and $\zeta(\xi) < \beta_A$ such that

$$\left(\bigcap_{i \in F(\xi)} U_i \right) \cap W_{\zeta(\xi)}^A \cap U_\xi = \emptyset.$$

We define

$$\varphi_A : A \setminus B_A \rightarrow \mathcal{P}_\omega^*(B_A) \times \beta_A$$

by the rule

$$\varphi_A(\xi) = (F(\xi), \zeta(\xi)),$$

and we set

$$\mathcal{P}_A = \{B_A\} \cup \{\varphi_A^{-1}(\{(F, \zeta)\}) : F \in \mathcal{P}_\omega^*(B_A), \zeta < \beta_A\}.$$

Then $\mathcal{P}_A$ is a partition of A , and since

$$|\mathcal{P}_\omega^*(B_A)| \leq \sup \{\omega, |B_A|\} < \alpha \text{ and } \beta_A < \alpha,$$

we have $|\mathcal{P}_A| < \alpha$. It follows from (the case $\beta = \alpha$ of) Lemma 1.1 that there is a family $\{A_\eta : \eta < \kappa\}$ of subsets of α such that

$$|A_\eta| = \alpha \quad \text{for } \eta < \kappa,$$

$$A_{\eta+1} \in \mathcal{P}_{A_\eta} \quad \text{for } \eta < \kappa, \text{ and}$$

$$A_{\eta'} \subset A_\eta \quad \text{for } \eta < \eta' < \kappa.$$

For $\eta < \kappa$ there are $\bar{F}(\eta) \in \mathcal{P}_\omega^*(B_{A_\eta})$, $\bar{\zeta}(\eta) < \beta_{A_\eta}$ such that

$$A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{(\bar{F}(\eta), \bar{\zeta}(\eta))\}).$$

For $\eta < \kappa$ we set

$$V_\eta = (\bigcap_{i \in \bar{F}(\eta)} U_i) \cap W_{\bar{\zeta}(\eta)}^{A_\eta},$$

and we note that $V_\eta \neq \emptyset$ (since $\bar{F}(\eta)$ is a finite subset of B_{A_η} and $\{U_i \cap W_{\bar{\zeta}(\eta)}^{A_\eta} : i \in B_{A_\eta}\}$ has the finite intersection property).

We claim further that

$$V_\eta \cap V_{\eta'} = \emptyset \quad \text{for } \eta < \eta' < \kappa.$$

Indeed since $\bar{F}(\eta') \neq \emptyset$, there is $\xi \in \bar{F}(\eta')$, and from

$$\xi \in \bar{F}(\eta') \subset A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{(\bar{F}(\eta), \bar{\zeta}(\eta))\})$$

we have

$$\varphi_{A_\eta}(\xi) = (\bar{F}(\eta), \bar{\zeta}(\eta))$$

and hence

$$\emptyset = (\bigcap_{i \in \bar{F}(\eta)} U_i) \cap W_{\bar{\zeta}(\eta)}^{A_\eta} \cap U_\xi = V_\eta \cap U_\xi;$$

then from $V_{\eta'} \subset U_\xi$ we have $V_\eta \cap V_{\eta'} = \emptyset$, as claimed.

It follows that $S(X) > \kappa$. This contradiction completes the proof.

5.2 Corollary. Let $\omega \leq \kappa \ll \alpha$ with α regular and let X be a space such that $S(X) \leq \kappa$. Then α is a precalibre for X .

Proof. Let $\{U_\xi : \xi < \alpha\}$ be a family of non-empty, open subsets of X , and let $\bar{\xi}$ be the ordinal number given by Lemma 5.1. With $\beta = 1$ and with

$$\{W_\zeta : \zeta < \beta\} = \{W_0\} = \{U_{\bar{\xi}}\},$$

it follows from Lemma 5.1 that there is $A \subset \alpha$ with $|A| = \alpha$ such that the family $\{U_\xi \cap U_{\bar{\xi}} : \xi \in A\}$ has the finite intersection property.

The proof is complete.

5.3 Remarks. (a) The power and usefulness of Lemma 5.1 will become apparent in 5.5 below, but for the moment we note that a simple (and similar) argument is sufficient to prove Corollary 5.2.

Suppose that α is not a precalibre for X , and let $\{U_\xi : \xi < \alpha\}$ be a set of non-empty, open subsets of X such that if $B \subset \alpha$ and $\{U_\xi : \xi \in B\}$ has the finite intersection property, then $|B| < \alpha$.

For $A \subset \alpha$ with $|A| = \alpha$ we set

$$\mathcal{B}_A = \{B \subset A : \{U_\xi : \xi \in B\} \text{ has the finite intersection property}\}.$$

Since the set $\mathcal{B}_A$ partially ordered by inclusion is inductive, there is a maximal element B_A of $\mathcal{B}_A$. We have $|B_A| < \alpha$, and since $\{\xi\} \in \mathcal{B}_A$ for all $\xi \in A$ we have $B_A \neq \emptyset$.

For $\xi \in A \setminus B_A$ it follows from the maximality of B_A that there is a finite, non-empty subset $F(\xi)$ of B_A such that

$$\left(\bigcap_{i \in F(\xi)} U_i\right) \cap U_\xi = \emptyset.$$

We define

$$\varphi_A : A \setminus B_A \rightarrow \mathcal{P}_\omega^*(B_A)$$

by the rule

$$\varphi_A(\xi) = F(\xi),$$

and we set

$$\mathcal{P}_A = \{B_A\} \cup \{\varphi_A^{-1}(\{F\}) : F \in \mathcal{P}_\omega^*(B_A)\}.$$

Then $\mathcal{P}_A$ is a partition of A , with $|\mathcal{P}_A| < \alpha$. It follows from (the case $\beta = \alpha$ of) Lemma 1.1 that there is a family $\{A_\eta : \eta < \kappa\}$ of subsets of α such that

$$\begin{aligned} |A_\eta| &= \alpha \quad \text{for } \eta < \kappa, \\ A_{\eta+1} &\in \mathcal{P}_{A_\eta} \quad \text{for } \eta < \kappa, \text{ and} \\ A_{\eta'} &\subset A_\eta \quad \text{for } \eta < \eta' < \kappa. \end{aligned}$$

For $\eta < \kappa$ there is $\bar{F}(\eta) \in \mathcal{P}_\omega^*(B_{A_\eta})$ such that

$$A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{\bar{F}(\eta)\}).$$

We set $V_\eta = \bigcap_{i \in \bar{F}(\eta)} U_i$ for $\eta < \kappa$ and we prove, just as in the proof of Lemma 5.1, that the family $\{V_\eta : \eta < \kappa\}$ is a faithfully indexed family of non-empty, pairwise disjoint open subsets of X . (Indeed $V_\eta \neq \emptyset$ since $\bar{F}(\eta)$ is a finite subset of B_{A_η} and $\{U_i : i \in B_{A_\eta}\}$ has the finite intersection property; and for $\eta < \eta' < \kappa$ there is $\xi \in \bar{F}(\eta')$, and from

$$\xi \in \bar{F}(\eta') \subset A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{\bar{F}(\eta)\})$$

we have $\varphi_{A_\eta}(\xi) = \bar{F}(\eta)$ and hence

$$V_\eta \cap V_{\eta'} \subset V_\eta \cap U_\xi = \emptyset.$$

This is a contradiction to the fact that $S(X) \leq \kappa$, and the proof is complete.

(b) The argument just given applies also to prove the following result. Here we use Lemma 1.2 instead of Lemma 1.1 with $\beta = \alpha$; and we replace κ throughout with α .

Theorem. Let α be a weakly compact cardinal and let X be a space such that $S(X) \leq \alpha$. Then α is a precalibre for X .

We use the following lemma in the proofs of Theorems 5.5 and 5.25.

5.4 Lemma. Let α be a singular cardinal, $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ a set of regular cardinals such that

$$\text{cf}(\alpha) < \alpha_{\sigma'} < \alpha_\sigma < \alpha \quad \text{for } \sigma' < \sigma < \text{cf}(\alpha), \text{ and}$$

$$\alpha = \sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma,$$

and let X be a space and $\{U_\xi : \xi < \alpha\}$ a set of non-empty open subsets of X . If there is a set $\{\bar{\xi}(\sigma) : \sigma < \text{cf}(\alpha)\}$ such that

$$\bar{\xi}(\sigma) \text{ is a representative of } \{U_\xi : \xi < \alpha_\sigma\} \text{ and}$$

$$\{U_{\bar{\xi}(\sigma)} : \sigma < \text{cf}(\alpha)\} \text{ has the finite intersection property,}$$

then there is $A \subset \alpha$ with $|A| = \alpha$ such that

$$\{U_\xi : \xi \in A\} \text{ has the finite intersection property.}$$

Proof. We define a family $\{A_\sigma : \sigma < \text{cf}(\alpha)\}$ such that

$$A_\sigma \subset \alpha_\sigma \quad \text{for } \sigma < \text{cf}(\alpha),$$

$$|A_\sigma| = \alpha_\sigma \quad \text{for } \sigma < \text{cf}(\alpha), \text{ and}$$

$$\{U_\xi : \xi \in \bigcup_{\sigma < \text{cf}(\alpha)} A_\sigma\} \cup \{U_{\bar{\xi}(\sigma)} : \sigma < \text{cf}(\alpha)\}$$

has the finite intersection property.

We proceed by recursion. We note that

$$|\mathcal{P}_\omega^*(\text{cf}(\alpha))| = \text{cf}(\alpha) < \alpha_0,$$

we set

$$\beta_0 = |\mathcal{P}_\omega^*(\text{cf}(\alpha))|,$$

and we let $\{W_\zeta : \zeta < \beta_0\}$ be an enumeration of

$$\{(\bigcap_{\tau \in F} U_{\xi(\tau)}) \cap U_{\xi(0)} : F \in \mathcal{P}_\omega^*(\text{cf}(\alpha))\}.$$

Then $\{W_\zeta : \zeta < \beta_0\}$ is a set of non-empty, open subsets of $U_{\xi(0)}$, and since $\xi(0)$ is a representative of $\{U_\xi : \xi < \alpha_0\}$ there is $A_0 \subset \alpha_0$ with $|A_0| = \alpha_0$ such that $\{U_\xi \cap W_\zeta : \xi \in A_0\}$ has the finite intersection property for all $\zeta < \beta_0$.

Now let $\sigma < \text{cf}(\alpha)$, and suppose that $A_{\sigma'}$ has been defined for $\sigma' < \sigma$ so that

$$\begin{aligned} A_{\sigma'} &\subset \alpha_{\sigma'}, \\ |A_{\sigma'}| &= \alpha_{\sigma'}, \text{ and} \\ \{U_\xi : \xi \in \bigcup_{\sigma' < \sigma} A_{\sigma'}\} \cup \{U_{\xi(\sigma)} : \sigma < \text{cf}(\alpha)\} \end{aligned}$$

has the finite intersection property.

Again we have $|\mathcal{P}_\omega^*(\text{cf}(\alpha))| = |\text{cf}(\alpha)| = \text{cf}(\alpha) < \alpha_\sigma$. Further, since

$$|\sigma| < \text{cf}(\alpha) < \alpha_\sigma \text{ and } \alpha_\sigma \text{ is regular}$$

we have

$$|\mathcal{P}_\omega^*(\bigcup_{\sigma' < \sigma} A_{\sigma'})| = |\bigcup_{\sigma' < \sigma} A_{\sigma'}| < \alpha_\sigma.$$

We set

$$\beta_\sigma = |\mathcal{P}_\omega^*(\text{cf}(\alpha)) \cup \mathcal{P}_\omega^*(\bigcup_{\sigma' < \sigma} A_{\sigma'})|,$$

we note that $\beta_\sigma < \alpha_\sigma$, and we let $\{W_\zeta : \zeta < \beta_\sigma\}$ be an enumeration of

$$\{(\bigcap_{\tau \in F} U_{\xi(\tau)}) \cap (\bigcap_{\xi \in G} U_\xi) \cap U_{\xi(\sigma)} : F \in \mathcal{P}_\omega^*(\text{cf}(\alpha)), G \in \mathcal{P}_\omega^*(\bigcup_{\sigma' < \sigma} A_{\sigma'})\}.$$

Then $\{W_\zeta : \zeta < \beta_\sigma\}$ is a set of non-empty, open subsets of $U_{\xi(\sigma)}$, and since $\xi(\sigma)$ is a representative of $\{U_\xi : \xi < \alpha_\sigma\}$, there is $A_\sigma \subset \alpha_\sigma$ with $|A_\sigma| = \alpha_\sigma$ such that $\{U_\xi \cap W_\zeta : \xi \in A_\sigma\}$ has the finite intersection property for all $\zeta < \beta_\sigma$.

The inductive definition of the family $\{A_\sigma : \sigma < \text{cf}(\alpha)\}$ is complete.

We set

$$A = \bigcup_{\sigma < \text{cf}(\alpha)} A_\sigma.$$

Then $|A| = \alpha$, and

$\{U_\xi : \xi \in A\} \cup \{U_{\xi(\sigma)} : \sigma < \text{cf}(\alpha)\}$ has the finite intersection property.

In particular

$\{U_\xi : \xi \in A\}$ has the finite intersection property,

as required.

The proof is complete.

5.5 Theorem. Let α be an (infinite) singular cardinal and let X be a space with precalibre $\text{cf}(\alpha)$. Let $S(X) \leq \kappa$, and suppose there is a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ such that

$$\begin{aligned} \alpha_\sigma &\text{ is a regular cardinal for } \sigma < \text{cf}(\alpha), \\ \kappa &\ll \alpha_\sigma \text{ for } \sigma < \text{cf}(\alpha), \\ \text{cf}(\alpha) &< \alpha_{\sigma'} < \alpha_\sigma < \alpha \text{ for } \sigma' < \sigma < \text{cf}(\alpha), \text{ and} \\ \alpha &= \sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma. \end{aligned}$$

Then X has precalibre α .

Proof. Let $\{U_\xi : \xi < \alpha\}$ be a set of non-empty, open subsets of X .

For $\sigma < \text{cf}(\alpha)$ the set $\{U_\xi : \xi < \alpha_\sigma\}$ is a set of non-empty, open subsets of X . Since $\omega \leq \kappa \ll \alpha_\sigma$ and α_σ is regular, it follows from Lemma 5.1 (with α_σ in place of α) that there is $\xi(\sigma) < \alpha_\sigma$ such that $\xi(\sigma)$ is a representative of $\{U_\xi : \xi < \alpha_\sigma\}$. Without loss of generality we assume, using the fact that X has precalibre $\text{cf}(\alpha)$, that $\{U_{\xi(\sigma)} : \sigma < \text{cf}(\alpha)\}$ has the finite intersection property. It then follows from Lemma 5.4 that there is $A \subset \alpha$ with $|A| = \alpha$ such that

$\{U_\xi : \xi \in A\}$ has the finite intersection property.

The proof is complete.

5.6 Theorem. Let $\omega \leq \kappa \ll \alpha$ and $\kappa \ll \text{cf}(\alpha)$ and let X be a space such that $S(X) \leq \kappa$. Then X has precalibre $\text{cf}(\alpha)$ and X has precalibre α .

Proof. It follows from Corollary 5.2 (with α replaced by $\text{cf}(\alpha)$) that X has precalibre $\text{cf}(\alpha)$. If α is singular, then from

$$\text{cf}(\kappa) \leq \kappa < \text{cf}(\alpha)$$

we have $\text{cf}(\kappa) \neq \text{cf}(\alpha)$ and hence from Theorem A.7 of the Appendix

there is a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ as in the statement of Theorem 5.5. The required conclusion then follows from Theorem 5.5.

5.7 Theorem. Let $\omega \leq \kappa \ll \alpha$ and let X be a space such that $S(X) \leq \kappa$ and X has precalibre $\text{cf}(\alpha)$. If either

- κ is a successor cardinal, or
- $\text{cf}(\kappa) \neq \text{cf}(\alpha)$, or
- α is a strong limit cardinal, or
- $S(X) < \kappa$,

then X has precalibre α .

Proof. It is enough to show that there is a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ as in the statement of Theorem 5.5. In the first three cases this follows from Theorem A.7 of the Appendix; in case $S(X) < \kappa$ we set $\kappa' = (S(X))^+$ and we note that $\omega \leq \kappa' \ll \alpha$ and $S(X) \leq \kappa'$ and κ' is a successor cardinal.

We note some consequences of Theorem 5.7.

5.8. Corollary. Let α be a strong limit cardinal and X a space. Then X has precalibre α if and only if X has precalibre $\text{cf}(\alpha)$.

In particular, if X is a compact Hausdorff space then X has calibre α if and only if X has calibre $\text{cf}(\alpha)$.

Proof. The first equivalence follows from Corollary 2.25 and Theorem 5.7, the second from Corollary 2.24.

5.9 Corollary. Assume the generalized continuum hypothesis. Let α be an infinite cardinal and X a space. Then X has precalibre α if and only if X has precalibre $\text{cf}(\alpha)$.

In particular, if X is a compact Hausdorff space then X has calibre α if and only if X has calibre $\text{cf}(\alpha)$.

Proof. If α is regular there is nothing to prove. If α is singular then α is a limit cardinal, hence (from the generalized continuum hypothesis) a strong limit cardinal, and the required conclusion follows from Corollary 5.8.

5.10 Corollary. Assume the generalized continuum hypothesis. Let α be an (infinite) cardinal with $\text{cf}(\alpha) > \omega$, let $\{X_i : i \in I\}$ be a set of

non-empty spaces, and $X = \prod_{i \in I} X_i$. Then α is a precalibre for X if and only if α is a precalibre for X_i for all $i \in I$.

In particular, if X_i is a compact Hausdorff space for all $i \in I$ then α is a calibre for X if and only if α is a calibre for X_i for all $i \in I$.

Proof. Of the following seven statements, each of the first six is equivalent to its successor: α is a precalibre for X ; $\text{cf}(\alpha)$ is a precalibre for X ; $\text{cf}(\alpha)$ is a precalibre for $G(X)$; $\text{cf}(\alpha)$ is a precalibre for $\prod_{i \in I} G(X_i)$; $\text{cf}(\alpha)$ is a precalibre for $G(X_i)$ for all $i \in I$; $\text{cf}(\alpha)$ is a precalibre for X_i for all $i \in I$; α is a precalibre for X_i for all $i \in I$. (The indicated equivalences are given respectively by 5.9; 2.22; 2.18(b) and 2.21; 3.7(a); 2.18(b); and 5.9.)

The second statement of the corollary follows from Corollary 2.24.

It follows from Corollary 3.6(a) that for α a singular cardinal with $\text{cf}(\alpha) > \omega$ the space $X = X_I$ of Theorem 4.4 is a product of (non-compact) Hausdorff spaces each with calibre α (hence with calibre $\text{cf}(\alpha)$) such that X has calibre $\text{cf}(\alpha)$ and X does not have calibre α . Thus the analogues of the second paragraphs of Corollaries 5.9 and 5.10 for non-compact spaces both fail (cf. Theorems 4.3 and 4.4).

The results above suggest some natural questions.

5.11 Problems. (a) Assume the generalized continuum hypothesis and let κ , β and γ be infinite cardinals with κ regular, $\text{cf}(\beta) < \kappa$, $\text{cf}(\gamma) < \kappa$, and $\gamma > \beta$. Is there a compact space X such that $S(X) = \kappa$ and X has calibre β^+ and X does not have calibre γ^+ ?

Even the following (simplest) case of this question is unsolved: Assuming the generalized continuum hypothesis, is there a compact c.c.c. space for which ω^+ is, and $(\aleph_\omega)^+$ is not, a calibre?

We note that from an answer (whether positive or negative) to (a) there will follow, assuming the generalized continuum hypothesis, a characterization (analogous to Corollary 4.7) of those classes of infinite cardinal numbers which can arise as the class of calibres of a compact Hausdorff space.

(b) Let $\omega \leq \kappa \leq \alpha$ with $\kappa = \text{cf}(\alpha)$ and let X be a space such that $S(X) = \kappa$ and X has precalibre $\text{cf}(\alpha)$. Does it follow (without addi-

tional set-theoretic assumptions, or hypotheses on α or κ) that X must have precalibre α ?

In connection with problem (b) we mention the following purely combinatorial question.

(c) Let $\omega \leq \kappa \ll \alpha$ and $\kappa \ll \text{cf}(\alpha)$ with κ regular. Does it follow that $\alpha \rightarrow (\alpha, \kappa)^2$?

(Pre-)calibres for P_{λ^+} -spaces and λ^+ -box products

In this section we indicate how several of the results above generalize (without additional assumptions) from spaces X to spaces of the form $X_{(\lambda^+)}$ and to λ^+ -box products (of the form $(X_I)_{\lambda^+}$). The arguments required in the two cases have many similarities (and are, in addition, similar to much of what has appeared in the proofs of 5.1–5.10). To avoid unnecessary repetition we give complete proofs below relating to the spaces $X_{(\lambda^+)}$ in question (cf. 5.12–5.16) and we give fewer details concerning the products $(X_I)_{\lambda^+}$ (cf. 5.17–5.21).

We begin with a generalization of Corollary 5.2.

5.12 Theorem. Let $\omega \leq \lambda < \kappa \ll \alpha$ with α and κ regular and let X be a compact Hausdorff space such that $S(X) \leq \kappa$. Then $X_{(\lambda^+)}$ has calibre α .

Proof. It is, by Corollary 2.24, sufficient to show that $X_{(\lambda^+)}$ has precalibre α . Let $\{U_\xi : \xi < \alpha\}$ be a set of non-empty, open subsets of $X_{(\lambda^+)}$. We assume without loss of generality, using Lemma 2.23, that for $\xi < \alpha$ there is a set $\{U_\xi(\zeta) : \zeta < \lambda\}$ of (non-empty) open subsets of X such that

$$U_\xi = \bigcap_{\zeta < \lambda} U_\xi(\zeta) = \bigcap_{\zeta < \lambda} \text{cl}_X U_\xi(\zeta),$$

$$\text{cl}_X U_\xi(\zeta) \text{ is compact for } \zeta < \lambda, \text{ and}$$

$$\{U_\xi(\zeta) : \zeta < \lambda\} \text{ is closed under finite intersections.}$$

Suppose that if $B \subset \alpha$ and $\{U_\xi : \xi \in B\}$ has the finite intersection property then $|B| < \alpha$, and for $A \subset \alpha$ with $|A| = \alpha$ set

$$\mathcal{B}_A = \{B \subset A : \{U_\xi : \xi \in B\} \text{ has the finite intersection property}\}.$$

Since the set $\mathcal{B}_A$ partially ordered by inclusion is inductive, there

is a maximal element B_A of $\mathcal{B}_A$. We have $|B_A| < \alpha$, and since $\{\xi\} \in \mathcal{B}_A$ for all $\xi \in A$ we have $B_A \neq \emptyset$.

For $\xi \in A \setminus B_A$ it follows from the maximality of B_A that there is a finite, non-empty subset $F(\xi)$ of B_A such that

$$\left(\bigcap_{i \in F(\xi)} U_i \right) \cap U_\xi = \emptyset.$$

From the conditions on the families $\{U_\xi(\zeta) : \zeta < \lambda\}$ it follows that there are $f_\xi \in \lambda^{F(\xi)}$ and $\zeta(\xi) < \lambda$ such that

$$\bigcap_{i \in F(\xi)} U_i(f_\xi(i)) \cap U_\xi(\zeta(\xi)) = \emptyset.$$

We define

$$\varphi_A : A \setminus B_A \rightarrow \mathcal{P}_\omega^*(B_A) \times \cup \{\lambda^F : F \in \mathcal{P}_\omega^*(B_A)\} \times \lambda$$

by the rule

$$\varphi_A(\xi) = (F(\xi), f_\xi, \zeta(\xi)),$$

and we set

$$\mathcal{P}_A = \{B_A\} \cup \{\varphi_A^{-1}(\{(F, f, \zeta)\}) : F \in \mathcal{P}_\omega^*(B_A), f \in \lambda^F, \zeta < \lambda\}.$$

Then $\mathcal{P}_A$ is a partition of A , and since $|B_A| < \alpha$ and $\lambda < \alpha$ we have $|\mathcal{P}_A| < \alpha$. It follows from (the case $\beta = \alpha$ of) Lemma 1.1 that there is a family $\{A_\eta : \eta < \kappa\}$ of subsets of α such that

$$\begin{aligned} |A_\eta| &= \alpha \quad \text{for } \eta < \kappa, \\ A_{\eta+1} &\in \mathcal{P}_{A_\eta} \quad \text{for } \eta < \kappa, \text{ and} \\ A_{\eta'} &\subset A_\eta \quad \text{for } \eta < \eta' < \kappa. \end{aligned}$$

For $\eta < \kappa$ there are $\bar{F}(\eta) \in \mathcal{P}_\omega^*(B_A)$ and $\bar{f}_\eta \in \lambda^{\bar{F}(\eta)}$ and $\bar{\zeta}(\eta) < \lambda$ such that

$$A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{\bar{F}(\eta), \bar{f}_\eta, \bar{\zeta}(\eta)\}).$$

We assume without loss of generality, using the fact that κ is regular and $\lambda < \kappa$, that there is $\bar{\zeta} < \lambda$ such that $\bar{\zeta}(\eta) = \bar{\zeta}$ for all $\eta < \kappa$.

We set

$$V_\eta = \left(\bigcap \{U_i(\bar{f}_\eta(i)) : i \in \bar{F}(\eta)\} \right) \cap \left(\bigcap \{U_i(\bar{\zeta}) : i \in \bar{F}(\eta)\} \right) \quad \text{for } \eta < \kappa.$$

Since V_η is the intersection of a finite number of open subsets of X , the set V_η is open in X . Further for $\eta < \kappa$ we have, using the fact that $\bar{F}(\eta)$ is a finite subset of B_{A_η} and that $\{U_i : i \in B_{A_\eta}\}$ has the finite

intersection property, that

$$V_\eta \supset \cap \{U_i : i \in \bar{F}(\eta)\} \neq \emptyset.$$

We claim finally that

$$V_\eta \cap V_{\eta'} = \emptyset \quad \text{for } \eta < \eta' < \kappa.$$

Indeed since $\bar{F}(\eta') \neq \emptyset$ there is $\xi \in \bar{F}(\eta')$, and from

$$\xi \in \bar{F}(\eta') \subset A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{\bar{F}(\eta), \bar{f}_\eta, \bar{\xi}\})$$

we have

$$\varphi_{A_\eta}(\xi) = (\bar{F}(\eta), \bar{f}_\eta, \bar{\xi})$$

and hence

$$\bigcap_{i \in \bar{F}(\eta)} U_i(\bar{f}_\eta(i)) \cap U_\xi(\bar{\xi}) = \emptyset;$$

then from $\bigcap_{i \in \bar{F}(\eta)} U_i(\bar{f}_\eta(i)) \supset V_\eta$ and $U_\xi(\bar{\xi}) \supset V_{\eta'}$, we have $V_\eta \cap V_{\eta'} = \emptyset$, as claimed.

It follows that $S(X) > \kappa$. This contradiction completes the proof.

5.13 Corollary. Let $\alpha \geq \omega$ and let X be a compact Hausdorff space such that $S(X) \leq \alpha^+$. Then $X_{(\alpha^+)}$ has calibre $(2^\alpha)^+$.

Proof. We have remarked in A.5(b) of the Appendix that $\alpha^+ \ll (2^\alpha)^+$. The statement then follows from Theorem 5.12 with λ, κ and α replaced by α, α^+ and $(2^\alpha)^+$ respectively.

We now turn to the case of singular cardinals. The first lemma generalizes Lemma 5.1.

5.14 Lemma. Let $\omega \leq \lambda < \kappa \ll \alpha$ with α and κ regular and let X be a compact Hausdorff space such that $S(X) \leq \kappa$. Let $\{U_\xi : \xi < \alpha\}$ be a set of non-empty, open subsets of $X_{(\lambda^+)}$ such that for $\xi < \alpha$ there is a set $\{U_\zeta(\zeta) : \zeta < \lambda\}$ of open subsets of X such that

$$U_\xi = \bigcap_{\zeta < \lambda} U_\zeta(\zeta) = \bigcap_{\zeta < \lambda} \text{cl}_X U_\zeta(\zeta),$$

$$\text{cl}_X U_\zeta(\zeta) \text{ is compact for } \zeta < \lambda, \text{ and}$$

$$\{U_\xi(\zeta) : \zeta < \lambda\} \text{ is closed under finite intersections.}$$

Then the set $\{U_\xi : \xi < \alpha\}$ has a representative.

Proof. We suppose the statement fails. For $A \subset \alpha$ with $|A| = \alpha$ we choose $\xi_A \in A$. There are

a cardinal $\beta_A < \alpha$, and

a set $\{W_v^A : v < \beta_A\}$ of non-empty, open (in $X_{(\lambda^+)}$) subsets of U_{ξ_A} , such that

if $B \subset A$ and $\{U_\xi \cap W_v^A : \xi \in B\}$ has the finite intersection property for all $v < \beta_A$, then $|B| < \alpha$.

We assume without loss of generality, using Lemma 2.23, that for $v < \beta_A$ there is a set $\{W_v^A(\zeta') : \zeta' < \lambda\}$ of (non-empty) open subsets of X such that

$$\begin{aligned} W_v^A &= \bigcap_{\zeta' < \lambda} W_v^A(\zeta') = \bigcap_{\zeta' < \lambda} \text{cl}_X W_v^A(\zeta'), \text{ and} \\ \text{cl}_X W_v^A(\zeta') &\text{ is compact for } \zeta' < \lambda, \text{ and} \\ \{W_v^A(\zeta') : \zeta' < \lambda\} &\text{ is closed under finite intersections.} \end{aligned}$$

For $A \subset \alpha$ with $|A| = \alpha$ we set

$$\mathcal{B}_A = \{B \subset A : \{U_\xi \cap W_v^A : \xi \in B\} \text{ has the finite intersection property for all } v < \beta_A\}.$$

Since the set $\mathcal{B}_A$ partially ordered by inclusion is inductive, there is a maximal element B_A of $\mathcal{B}_A$. We have $|B_A| < \alpha$, and since $\{\xi_A\} \in \mathcal{B}_A$ we have $B_A \neq \emptyset$.

For $\xi \in A \setminus B_A$ it follows from the maximality of B_A that there are a finite, non-empty subset $F(\xi)$ of B_A and $v(\xi) < \beta_A$ such that

$$\left(\bigcap_{i \in F(\xi)} U_i \right) \cap W_{v(\xi)}^A \cap U_\xi = \emptyset.$$

From the conditions on the families $\{U_\xi(\zeta) : \zeta < \lambda\}$ and $\{W_v^A(\zeta') : \zeta' < \lambda\}$ it follows that there are $f_\xi \in \lambda^{F(\xi)}$ and $\zeta(\xi), \zeta'(\xi) < \lambda$ such that

$$\left(\bigcap_{i \in F(\xi)} U_i(f_\xi(i)) \right) \cap W_{v(\xi)}^A(\zeta'(\xi)) \cap U_\xi(\zeta(\xi)) = \emptyset.$$

We define

$$\varphi_A : A \setminus B_A \rightarrow \mathcal{P}_\omega^*(B_A) \times \cup \{\lambda^F : F \in \mathcal{P}_\omega^*(B_A)\} \times \beta_A \times \lambda \times \lambda$$

by the rule

$$\varphi_A(\xi) = (F(\xi), f_\xi, v(\xi), \zeta'(\xi), \zeta(\xi)),$$

and we set

$$\begin{aligned} \mathcal{P}_A &= \{B_A\} \cup \{\varphi_A^{-1}(\{(F, f, v, \zeta', \zeta)\}) : \\ &\quad F \in \mathcal{P}_\omega^*(B_A), f \in \lambda^F, v < \beta_A, \zeta', \zeta < \lambda\}. \end{aligned}$$

Then $\mathcal{P}_A$ is a partition of A , and since $|B_A| < \alpha$, $\beta_A < \alpha$ and $\lambda < \alpha$

we have $|\mathcal{P}_A| < \alpha$. It follows from (the case $\beta = \alpha$ of) Lemma 1.1 that there is a family $\{A_\eta : \eta < \kappa\}$ of subsets of α such that

$$\begin{aligned} |A_\eta| &= \alpha \quad \text{for } \eta < \kappa, \\ A_{\eta+1} &\in \mathcal{P}_{A_\eta} \quad \text{for } \eta < \kappa, \text{ and} \\ A_{\eta'} &\subset A_\eta \quad \text{for } \eta < \eta' < \kappa. \end{aligned}$$

For $\eta < \kappa$ there are $\bar{F}(\eta) \in \mathcal{P}_\omega^*(B_{A_\eta})$, $\bar{f}_\eta \in \lambda^{\bar{F}(\eta)}$, $\bar{v}(\eta) < \beta_{A_\eta}$, $\bar{\zeta}'(\eta) < \lambda$ and $\bar{\zeta}(\eta) < \lambda$ such that

$$A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{(\bar{F}(\eta), \bar{f}_\eta, \bar{v}(\eta), \bar{\zeta}'(\eta), \bar{\zeta}(\eta))\}).$$

We assume without loss of generality, using the fact that κ is regular and $\lambda < \kappa$, that there are $\bar{\zeta}', \bar{\zeta} < \lambda$ such that $\bar{\zeta}'(\eta) = \bar{\zeta}'$ and $\bar{\zeta}(\eta) = \bar{\zeta}$ for all $\eta < \kappa$.

We set

$$V_\eta = (\cap \{U_i(\bar{f}_\eta(i)) : i \in \bar{F}(\eta)\}) \cap W_{\bar{v}(\eta)}^{A_\eta}(\bar{\zeta}') \cap (\cap \{U_i(\bar{\zeta}) : i \in \bar{F}(\eta)\}).$$

Since V_η is the intersection of a finite number of open subsets of X , the set V_η is open in X . Further for $\eta < \kappa$ we have, using the facts that $\bar{F}(\eta)$ is a finite subset of B_{A_η} and $\{U_i \cap W_{\bar{v}(\eta)}^{A_\eta} : i \in B_{A_\eta}\}$ has the finite intersection property and

$$V_\eta \supset (\cap_{i \in \bar{F}(\eta)} U_i) \cap W_{\bar{v}(\eta)}^{A_\eta},$$

that $V_\eta \neq \emptyset$. We claim finally that

$$V_\eta \cap V_{\eta'} = \emptyset \quad \text{for } \eta < \eta' < \kappa.$$

Indeed since $\bar{F}(\eta') \neq \emptyset$ there is $\xi \in \bar{F}(\eta')$, and from

$$\xi \in \bar{F}(\eta') \subset A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{(\bar{F}(\eta), \bar{f}_\eta, \bar{v}(\eta), \bar{\zeta}', \bar{\zeta})\})$$

we have

$$\varphi_{A_\eta}(\xi) = (\bar{F}(\eta), \bar{f}_\eta, \bar{v}(\eta), \bar{\zeta}', \bar{\zeta})$$

and hence

$$\cap_{i \in \bar{F}(\eta)} U_i(\bar{f}_\eta(i)) \cap W_{\bar{v}(\eta)}^{A_\eta}(\bar{\zeta}') \cap U_\xi(\bar{\zeta}) = \emptyset;$$

then from $\cap_{i \in \bar{F}(\eta)} U_i(\bar{f}_\eta(i)) \cap W_{\bar{v}(\eta)}^{A_\eta}(\bar{\zeta}') \supset V_\eta$ and $U_\xi(\bar{\zeta}) \supset V_{\eta'}$ we have $V_\eta \cap V_{\eta'} = \emptyset$, as claimed.

It follows that $S(X) > \kappa$. This contradiction completes the proof.

5.15 Theorem. Let $\omega \leq \lambda < \kappa \leq \alpha$ with κ regular and α singular

and let X be a compact, Hausdorff space such that $S(X) \leq \kappa$ and $X_{(\lambda^+)}$ has calibre $\text{cf}(\alpha)$. If there is a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ such that

$$\begin{aligned} \alpha_\sigma &\text{ is a regular cardinal for } \sigma < \text{cf}(\alpha), \\ \kappa &\ll \alpha_\sigma \text{ for } \sigma < \text{cf}(\alpha), \\ \text{cf}(\alpha) &< \alpha_{\sigma'} < \alpha_\sigma < \alpha \text{ for } \sigma' < \sigma < \text{cf}(\alpha), \text{ and} \\ \alpha &= \sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma, \end{aligned}$$

then $X_{(\lambda^+)}$ has calibre α .

The proof is similar to that of Theorem 5.4; we here use Lemma 5.14 in place of Lemma 5.1. (We note that in order to show that for every set $\{U_\xi : \xi < \alpha\}$ of non-empty, open subsets of $X_{(\lambda^+)}$ there is $A \subset \alpha$ with $|A| = \alpha$ such that $\{U_\xi : \xi \in A\}$ has the finite intersection property, it is by Lemma 2.23 sufficient to consider the case that for $\xi < \alpha$ the set U_ξ has the form $U_\xi = \cap_{\zeta < \lambda} U_\xi(\zeta)$ with $U_\xi(\zeta)$ as in the statement of Lemma 5.14.)

5.16 Theorem. Let $\omega \leq \lambda < \kappa \ll \alpha$ and $\kappa \ll \text{cf}(\alpha)$ with κ regular and let X be a compact, Hausdorff space such that $S(X) \leq \kappa$. Then $X_{(\lambda^+)}$ has calibre α .

Proof. It follows from Theorem 5.10 (applied to $\text{cf}(\alpha)$) that $X_{(\lambda^+)}$ has calibre $\text{cf}(\alpha)$. Since $\kappa \ll \alpha$ and $\text{cf}(\kappa) \leq \kappa < \text{cf}(\alpha)$ there is (for α singular) by Theorem A.7 of the Appendix a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ satisfying the conditions of Theorem 5.15. Hence $X_{(\lambda^+)}$ has calibre α .

We recall from Chapter 3 that for $\kappa \geq \omega$ and $\{X_i : i \in I\}$ a set of spaces, the product $\prod_{i \in I} X_i$ with the κ -box topology is denoted $(\prod_{i \in I} X_i)_\kappa$ or $(X_I)_\kappa$. It is not required that $(X_I)_\kappa$ is a P_κ -space.

5.17 Theorem. Let $\omega \leq \lambda < \kappa \ll \alpha$ with α and κ regular, and let $\{X_i : i \in I\}$ be a set of spaces such that $S(X_i) \leq \kappa$ for $i \in I$. Then $(X_I)_{\lambda^+}$ has precalibre α .

Proof. From the statement and theorem following (the proof of) Theorem 3.6 it is enough to consider the case $I = \lambda$. If the lemma is false in this case then there is a set $\{U^\xi : \xi < \alpha\}$ of non-empty basic open subsets of $(\prod_{\zeta < \lambda} X_\zeta)_{\lambda^+}$ such that if $B \subset \alpha$ and $\{U^\xi : \xi \in B\}$ has the finite intersection property, then $|B| < \alpha$. We write $U^\xi = \prod_{\zeta < \lambda} U_\zeta^\xi$ with U_ζ^ξ a (non-empty) open subset of X_ζ for $\xi < \alpha, \zeta < \lambda$.

For $A \subset \alpha$ with $|A| = \alpha$ we set

$$\mathcal{B}_A = \{B \subset A : \{U^\xi : \xi \in B\} \text{ has the finite intersection property}\}.$$

Since the set $\mathcal{B}_A$ partially ordered by inclusion is inductive, there is a maximal element B_A of $\mathcal{B}_A$. We have $|B_A| < \alpha$, and since $\{\xi\} \in \mathcal{B}_A$ for all $\xi \in A$ we have $B_A \neq \emptyset$.

For $\xi \in A \setminus B_A$ it follows from the maximality of B_A that there is a finite, non-empty subset $F(\xi)$ of B_A such that

$$\left(\bigcap_{i \in F(\xi)} U^i\right) \cap U^\xi = \emptyset;$$

it follows from Lemma 3.1 that there is $\zeta(\xi) < \lambda$ such that

$$\left(\bigcap_{i \in F(\xi)} U^i_{\zeta(\xi)}\right) \cap U^\xi_{\zeta(\xi)} = \emptyset.$$

We define

$$\varphi_A : A \setminus B_A \rightarrow \mathcal{P}_\omega^*(B_A) \times \lambda$$

by the rule

$$\varphi_A(\xi) = (F(\xi), \zeta(\xi)),$$

and we set

$$\mathcal{P}_A = \{B_A\} \cup \{\varphi_A^{-1}(\{(F, \zeta)\}) : F \in \mathcal{P}_\omega^*(B_A), \zeta < \lambda\}.$$

Then $\mathcal{P}_A$ is a partition of A , with $|\mathcal{P}_A| < \alpha$. It follows from (the case $\beta = \alpha$ of) Lemma 1.1 that there is a family $\{A_\eta : \eta < \kappa\}$ of subsets of α such that

$$\begin{aligned} |A_\eta| &= \alpha \quad \text{for } \eta < \kappa, \\ A_{\eta+1} &\in \mathcal{P}_{A_\eta} \quad \text{for } \eta < \kappa, \text{ and} \\ A_{\eta'} &\subset A_\eta \quad \text{for } \eta < \eta' < \kappa. \end{aligned}$$

For $\eta < \kappa$ there are $\bar{F}(\eta) \in \mathcal{P}_\omega^*(B_{A_\eta})$ and $\bar{\zeta}(\eta) < \lambda$ such that

$$A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{(\bar{F}(\eta), \bar{\zeta}(\eta))\}).$$

We assume without loss of generality, using the fact that κ is regular and $\lambda < \kappa$, that there is $\bar{\zeta} < \lambda$ such that $\bar{\zeta}(\eta) = \bar{\zeta}$ for all $\eta < \kappa$.

We set

$$V_\eta = \bigcap_{i \in \bar{F}(\eta)} U^i_{\bar{\zeta}}.$$

Since V_η is the intersection of a finite number of open subsets of $X_{\bar{\zeta}}$, the set V_η is open in $X_{\bar{\zeta}}$. Further for $\eta < \kappa$ we have, using the

fact that $\bar{F}(\eta)$ is a finite subset of B_{A_η} and that $\{U^i : i \in B_{A_\eta}\}$ has the finite intersection property, that $\bigcap_{i \in \bar{F}(\eta)} U^i \neq \emptyset$ and hence $V_\eta \neq \emptyset$. We claim finally that

$$V_\eta \cap V_{\eta'} = \emptyset \quad \text{for } \eta < \eta' < \kappa.$$

Indeed since $\bar{F}(\eta') \neq \emptyset$ there is $\xi \in \bar{F}(\eta')$, and from

$$\xi \in \bar{F}(\eta') \subset A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{(\bar{F}(\eta), \bar{\xi})\})$$

we have

$$\varphi_{A_\eta}(\xi) = (\bar{F}(\eta), \bar{\xi})$$

and hence

$$\left(\bigcap_{i \in \bar{F}(\eta)} U_\xi^i \right) \cap U_\xi^\xi = \emptyset;$$

then from $\bigcap_{i \in \bar{F}(\eta)} U_\xi^i = V_\eta$ and $U_\xi^\xi \supset V_{\eta'}$ we have $V_\eta \cap V_{\eta'} = \emptyset$, as claimed.

It follows that $S(X_\xi) > \kappa$. This contradiction completes the proof.

5.18 Corollary. Let $\alpha \geq \omega$ and let $\{X_i : i \in I\}$ be a set of spaces such that $S(X_i) \leq \alpha^+$ for $i \in I$. Then $(\prod_{i \in I} X_i)_{\alpha^+}$ has precalibre $(2^\alpha)^+$.

Proof. We have remarked in A.5(b) of the Appendix that $\alpha^+ \ll (2^\alpha)^+$. The statement then follows from Theorem 5.17 with λ, κ and α replaced by α, α^+ and $(2^\alpha)^+$, respectively.

5.19 Lemma. Let $\omega \leq \lambda < \kappa \ll \alpha$ with α and κ regular and let $\{X_\zeta : \zeta < \lambda\}$ be a set of spaces such that $S(X_\zeta) \leq \kappa$ for $\zeta < \lambda$. Then every set $\{U^\xi : \xi < \alpha\}$ of non-empty, open subsets of $(\prod_{\zeta < \lambda} X_\zeta)_{\lambda^+}$ has a representative.

Proof. If the statement fails then for $A \subset \alpha$ with $|A| = \alpha$ there are $\xi_A \in A$, a cardinal $\beta_A < \alpha$, and a set $\{W_v^A : v < \beta_A\}$ of non-empty (basic) open subsets of U^{ξ_A} such that

if $B \subset A$ and $\{U^\xi \cap W_v^A : \xi \in B\}$ has the finite intersection property for all $v < \beta_A$, then $|B| < \alpha$.

We write

$$U^\xi = \prod_{\zeta < \lambda} U^\xi(\zeta) \quad \text{for } \xi < \alpha, \text{ and}$$

$$W_v^A = \prod_{\zeta < \lambda} W_v^A(\zeta) \quad \text{for } v < \beta_A.$$

For $A \subset \alpha$ with $|A| = \alpha$ we set

$\mathcal{B}_A = \{B \subset A : \{U^\xi \cap W_v^A : \xi \in B\} \text{ has the finite intersection property for all } v < \beta_A\}$.

Since the set $\mathcal{B}_A$ partially ordered by inclusion is inductive, there is a maximal element B_A of $\mathcal{B}_A$. We have $|B_A| < \alpha$, and since $\{\xi_A\} \in \mathcal{B}_A$ we have $B_A \neq \emptyset$.

For $\xi \in A \setminus B_A$ it follows from the maximality of B_A that there are a finite, non-empty subset $F(\xi)$ of B_A and $v(\xi) < \beta_A$ such that

$$\left(\bigcap_{i \in F(\xi)} U^i \right) \cap W_{v(\xi)}^A \cap U^\xi = \emptyset;$$

hence there is $\zeta(\xi) < \lambda$ such that

$$\left(\bigcap_{i \in F(\xi)} U^i(\zeta(\xi)) \right) \cap W_{v(\xi)}^A(\zeta(\xi)) \cap U^{\zeta(\xi)} = \emptyset.$$

We define

$$\varphi_A : A \setminus B_A \rightarrow \mathcal{P}_\omega^*(B_A) \times \beta_A \times \lambda$$

by the rule

$$\varphi_A(\xi) = (F(\xi), v(\xi), \zeta(\xi)),$$

and we set

$$\mathcal{P}_A = \{B_A\} \cup \{\varphi_A^{-1}(\{(F, v, \zeta)\}) : F \in \mathcal{P}_\omega^*(B_A), v < \beta_A, \zeta < \lambda\}.$$

The proof concludes along lines by now familiar to the reader: there is a set $\{A_\eta : \eta < \kappa\}$ of subsets of α as in the conclusion of (the case $\beta = \alpha$ of) Lemma 1.1, for $\eta < \kappa$ there are $\bar{F}(\eta) \in \mathcal{P}_\omega^*(B_{A_\eta})$, $\bar{v}(\eta) < \beta_{A_\eta}$ and $\bar{\zeta}(\eta) < \lambda$ such that

$$A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{(\bar{F}(\eta), \bar{v}(\eta), \bar{\zeta}(\eta))\});$$

we assume without loss of generality that there is $\bar{\zeta} < \lambda$ such that $\bar{\zeta}(\eta) = \bar{\zeta}$ for all $\eta < \kappa$, we set

$$V_\eta = \left(\bigcap_{i \in \bar{F}(\eta)} U^i(\bar{\zeta}) \right) \cap W_{\bar{v}(\eta)}^{A_\eta}(\bar{\zeta}),$$

and we show as in earlier proofs that the (indexed) family $\{V_\eta : \eta < \kappa\}$ is a cellular family in $X_{\bar{\zeta}}$. This contradicts the assumption $S(X_{\bar{\zeta}}) \leq \kappa$.

5.20 Theorem. Let $\omega \leq \lambda < \kappa \ll \alpha$ with κ regular and α singular and let $\{X_i : i \in I\}$ be a set of spaces such that $S(X_i) \leq \kappa$ for $i \in I$ and $(X_i)_{i \in I}$ has precalibre $\text{cf}(\alpha)$. If there is a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ such

that

$$\begin{aligned} \alpha_\sigma &\text{ is a regular cardinal for } \sigma < \text{cf}(\alpha), \\ \kappa &\leq \alpha_\sigma \text{ for } \sigma < \text{cf}(\alpha), \\ \text{cf}(\alpha) &< \alpha_{\sigma'} < \alpha_\sigma < \alpha \text{ for } \sigma' < \sigma < \text{cf}(\alpha), \text{ and} \\ \alpha &= \sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma, \end{aligned}$$

then $(X_I)_{\lambda^+}$ has precalibre α .

From the theorem following (the proof of) Theorem 3.6 it is enough to consider the case $I = \lambda$. The proof is now similar to that of Theorem 5.5; we use Lemma 5.19 in place of Lemma 5.1.

5.21 Theorem. Let $\omega \leq \lambda < \kappa \leq \alpha$ and $\kappa \leq \text{cf}(\alpha)$ with κ regular, and let $\{X_i : i \in I\}$ be a set of spaces such that $S(X_i) \leq \kappa$ for $i \in I$. Then $(X_I)_{\lambda^+}$ has precalibre α .

Proof. It follows from Theorem 5.17 (with α replaced by $\text{cf}(\alpha)$) that $(X_I)_{\lambda^+}$ has precalibre $\text{cf}(\alpha)$. There is (for α singular), by Theorem A.7 of the Appendix, a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ satisfying the conditions of Theorem 5.20. Hence $(X_I)_{\lambda^+}$ has precalibre α .

Calibres of compact spaces: the ‘exceptional’ case

We recall from Appendix A that for $\alpha \geq \omega$, $\kappa > 2$ the cardinal numbers $\sqrt[\kappa]{\alpha}$, $\sqrt[\lambda]{\alpha}$ are defined by the relations

$$\begin{aligned} \sqrt[\kappa]{\alpha} &= \min\{\beta : \beta^\kappa \geq \alpha\}, \\ \sqrt[\lambda]{\alpha} &= \min\{\sqrt[\lambda]{\alpha} : 1 < \lambda < \kappa\}. \end{aligned}$$

5.22 Lemma. Let α and κ be regular cardinals with $2 < \sqrt[\kappa]{\alpha} < \alpha$, let X be a space with $S(X) \leq \kappa$, and let $\{U_\xi : \xi < \alpha\}$ be a set of non-empty, open subsets of X . Then there is $A \subset \alpha$ with $|A| = \alpha$ such that: if $C \subset A$ and $|C| \leq \kappa^\kappa$ and $\{U_\xi : \xi \in C\}$ has the finite intersection property, then

$|\{\xi \in A : \{U_\xi : \xi \in C \cup \{\xi\}\} \text{ has the finite intersection property}\}| = \alpha$.

Proof. We suppose the statement fails. For $A \subset \alpha$ with $|A| = \alpha$ there are $B_A \subset A$ with $|B_A| < \alpha$ and $C_A \subset A$ with $|C_A| \leq \kappa^\kappa$ such that $\{U_\xi : \xi \in C_A\}$ has the finite intersection property, and

for $\xi \in A \setminus (B_A \cup C_A)$ there is a finite, non-empty subset $F(\xi)$ of C_A such that $(\bigcap_{\zeta \in F(\xi)} U_\zeta) \cap U_\xi = \emptyset$.

We define

$$\varphi_A : A \setminus (B_A \cup C_A) \rightarrow \mathcal{P}_\omega^*(C_A)$$

by the rule

$$\varphi_A(\xi) = F(\xi),$$

and we set

$$\mathcal{P}_A = \{B_A \cup C_A\} \cup \{\varphi_A^{-1}(\{F\}) : F \in \mathcal{P}_\omega^*(C_A)\}.$$

Then $\mathcal{P}_A$ is a partition of α such that $|\mathcal{P}_A| \leq \kappa^\kappa$. We set

$$\beta = (\kappa^\kappa)^+.$$

It follows from Theorem A.11 of the Appendix that $\kappa \ll \beta$ and $\beta \leq \alpha$, and from Lemma 1.1 it then follows that there is a family $\{A_\eta : \eta < \kappa\}$ of subsets of α such that

$$\begin{aligned} |A_\eta| &= \alpha \quad \text{for } \eta < \kappa, \\ A_{\eta+1} &\in \mathcal{P}_{A_\eta} \quad \text{for } \eta < \kappa, \text{ and} \\ A_{\eta'} &\subset A_\eta \quad \text{for } \eta < \eta' < \kappa. \end{aligned}$$

For $\eta < \kappa$ there is $\bar{F}(\eta) \in \mathcal{P}_\omega^*(C_{A_\eta})$ such that

$$A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{\bar{F}(\eta)\}).$$

For $\eta < \kappa$ we set

$$V_\eta = \bigcap_{i \in \bar{F}(\eta)} U_i,$$

and we note that since $\bar{F}(\eta)$ is a finite subset of C_{A_η} and $\{U_i : i \in C_{A_\eta}\}$ has the finite intersection property, the open set V_η is non-empty. We claim finally that

$$V_\eta \cap V_{\eta'} = \emptyset \quad \text{for } \eta < \eta' < \kappa.$$

Indeed since $\bar{F}(\eta') \neq \emptyset$ there is $\xi \in \bar{F}(\eta')$, and from

$$\xi \in \bar{F}(\eta') \subset A_{\eta+1} = \varphi_{A_\eta}^{-1}(\{\bar{F}(\eta)\})$$

we have

$$\varphi_{A_\eta}(\xi) = \bar{F}(\eta)$$

and hence

$$V_\eta \cap V_{\eta'} \subset \left(\bigcap_{i \in \bar{F}(\eta)} U_i \right) \cap U_\xi = \emptyset.$$

It follows that $S(X) > \kappa$. This contradiction completes the proof.

We have seen in Theorem 2.1(h) for $\alpha \geq \omega$ that a space with calibre $(\alpha, 2)$ has calibre (α, n) for $0 < n < \omega$. We note now that in fact such a space has calibre (α, ω, n) for $0 < n < \omega$.

5.23 Lemma. Let $\alpha \geq \omega$ and let X be a space such that $S(X) \leq \alpha$.

(a) If α is regular and $\{U_\xi : \xi < \alpha\}$ is a set of non-empty, open subsets of X then there is $\zeta < \alpha$ such that

$$|\{\xi < \alpha : U_\xi \cap U_\zeta \neq \emptyset\}| = \alpha;$$

(b) X has precalibre (α, ω) .

Proof. (a) We assume the result fails and we define $\{A(\eta) : \eta < \alpha\} \subset \mathcal{P}(\alpha)$ and $\{\xi_\eta : \eta < \alpha\} \subset \alpha$ such that

- (i) $A(0) = \alpha, \xi_0 \in A(0)$;
- (ii) $A(\eta) = \{\xi < \alpha : U_\xi \cap U_{\xi_{\eta'}} = \emptyset \text{ for all } \eta' < \eta\}$ for $\eta < \alpha$; and
- (iii) $\xi_\eta \in A(\eta)$ for $\eta < \alpha$.

We proceed by transfinite recursion. We define $A(0)$ and ξ_0 by (i), and $A(\eta)$ and ξ_η for $0 < \eta < \alpha$ by (ii) and (iii). We note that $|A(\eta)| = \alpha$ for $\eta < \alpha$, since α is regular and

$$|\{\xi < \alpha : \text{there is } \eta' < \eta \text{ such that } U_\xi \cap U_{\xi_{\eta'}} \neq \emptyset\}| < \alpha.$$

It is clear that $U_{\xi_{\eta'}} \cap U_{\xi_\eta} = \emptyset$ for $\eta' < \eta < \alpha$, a contradiction.

(b) Suppose first that α is regular and let $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X)$; we show there is $A \subset \alpha$ with $|A| = \omega$ such that $\{U_\xi : \xi \in A\}$ has the finite intersection property. From part (a) there are $\xi_0 < \alpha$ and $A_0 \subset \alpha$ with $|A_0| = \alpha$ such that

$$\{U_\xi \cap U_{\xi_0} : \xi \in A_0\} \subset \mathcal{T}^*(X).$$

It is then simple using part (a) to define by recursion (for $0 < n < \omega$) an ordinal number $\xi_n \in A_{n-1}$ and $A_n \subset A_{n-1}$ with $|A_n| = \alpha$ such that $\xi_n > \xi_{n-1}$ and

$$\{U_\xi \cap U_{\xi_0} \cap U_{\xi_1} \cap \dots \cap U_{\xi_n} : \xi \in A_n\} \subset \mathcal{T}^*(X).$$

We set $A = \{\xi_n : n < \omega\}$; it is clear that A is as required.

If α is singular we assume without loss of generality (using 2.18–2.20) that X is a compact, Hausdorff space, we assume further that X is an infinite space (since otherwise $|\mathcal{T}^*(X)| < \omega$ and the statement

to be proved is obvious), we set $\beta = S(X)$, we recall from Theorem 2.14 that β is a regular cardinal, and we note from the preceding paragraph (with β replacing α) that X has precalibre (β, ω) ; since $\alpha \geq \beta$ it then follows that X has precalibre (α, ω) , as required.

The proof is complete.

It follows from Corollary 2.24 and Lemma 5.23 that if α is an infinite cardinal and X is a locally compact, Hausdorff space such that $S(X) \leq \alpha$, then X has calibre (α, ω) . We note now in passing that in this statement the condition that X is locally compact cannot be omitted.

5.24 Theorem. For $\alpha > \omega$ there is a space X such that X has calibre $(\alpha, 2)$ and X does not have calibre (α, ω) .

Proof. Set $Y = \{0, 1\}^\alpha$ and

$$X = \{x \in Y : |\{\xi < \alpha : x_\xi \neq 0\}| < \omega\}.$$

For $J \in \mathcal{P}_\omega^*(\alpha)$ the (finite) space $\prod_{\xi \in J} \{0, 1\}_\xi$ has calibre $(\omega^+, 2)$; hence Y has calibre $(\omega^+, 2)$ by Theorem 3.6(a). Since X is dense in Y , the space X has calibre $(\omega^+, 2)$; thus X has calibre $(\alpha, 2)$ by Theorem 2.1(a).

To prove that X does not have calibre (α, ω) it is enough to define $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X)$ such that $\bigcap_{\xi \in A} U_\xi = \emptyset$ for all $A \in [\alpha]^\omega$. For this we set

$$U_\xi = \{x \in X : x_\xi = 1\} \quad \text{for } \xi < \alpha.$$

5.25 Theorem. Let α and κ be (infinite) regular cardinals such that $\kappa \leq \alpha$, with $\kappa < \alpha$ in case α is a strong limit cardinal, and let X be a space such that $S(X) \leq \kappa$. Then X has precalibre $(\alpha, \omega \cdot \sqrt[\kappa]{\alpha})$.

Proof. We consider three cases.

Case 1. $\sqrt[\kappa]{\alpha} = 2$. Then $\omega \cdot \sqrt[\kappa]{\alpha} = \omega$ and the required conclusion follows from Lemma 5.23(b).

Case 2. $\sqrt[\kappa]{\alpha} = \alpha$. Since α is a strong limit cardinal and $\kappa < \alpha$ we have $\kappa \ll \alpha$. It follows from Corollary 5.2 that X has precalibre α , as required.

Case 3. $2 < \sqrt[\kappa]{\alpha} < \alpha$. We set $\beta = \sqrt[\kappa]{\alpha}$, we choose λ such that $\lambda < \kappa$ and $\beta = \sqrt[\lambda]{\alpha}$, and we note from Theorem A.9 of the Appendix

that

$$\text{cf}(\beta) = \text{cf}(\sqrt[\lambda]{\alpha}) \leq \lambda < \kappa.$$

We claim next that $\kappa \ll \beta$. If there are $\gamma < \beta$ and $\mu < \kappa$ such that $\gamma^\mu \geq \beta$, then from $\beta^\lambda \geq \alpha$ we have $\gamma^\mu \cdot \lambda \geq \alpha$ and hence (since $\mu \cdot \lambda < \kappa$)

$$\beta = \sqrt[\kappa]{\alpha} \leq \gamma < \beta,$$

a contradiction. Since $2 < \beta$ the relation $\beta < \kappa$ fails (otherwise $2^\beta < \beta$) and hence $\kappa \leq \beta$; since the three conditions

$$\text{cf}(\beta) < \kappa \leq \beta, \kappa \text{ is regular}, \kappa = \beta$$

are incompatible, we have $\kappa < \beta$. The proof that $\kappa \ll \beta$ is complete.

It now follows from Theorem A.7 of the Appendix (applied to the singular cardinal β) that there is a set $\{\beta_\sigma : \sigma < \text{cf}(\beta)\}$ such that

β_σ is a regular cardinal for $\sigma < \text{cf}(\beta)$,

$\kappa \ll \beta_0$ for $\sigma < \text{cf}(\beta)$,

$\text{cf}(\beta) < \beta_{\sigma'} < \beta_\sigma < \beta$ for $\sigma' < \sigma < \text{cf}(\beta)$, and

$$\beta = \sum_{\sigma < \text{cf}(\beta)} \beta_\sigma.$$

Now let $\{U_\xi : \xi < \alpha\}$ be a set of non-empty, open subsets of X , and let A be the subset of α (with $|A| = \alpha$) given for this family by Lemma 5.22. We define $\{B_\sigma : \sigma < \text{cf}(\beta)\}, \{\xi(\sigma) : \sigma < \text{cf}(\beta)\}$ such that

- (i) $B_\sigma \subset A$ and $|B_\sigma| = \beta_\sigma$ for $\sigma < \text{cf}(\beta)$,
- (ii) $\xi(\sigma)$ is a representative of $\{U_\xi : \xi \in B_\sigma\}$, and
- (iii) $\{U_{\xi(\sigma)} : \sigma < \text{cf}(\beta)\}$ has the finite intersection property.

We proceed by recursion. We set $B_0 = \beta_0$ and, using Lemma 5.1 (applied to B_0 in place of α), we choose a representative $\xi(0)$ of $\{U_\xi : \xi \in B_0\}$.

Now let $\sigma < \text{cf}(\beta)$ and suppose for $\sigma' < \sigma$ that $B_{\sigma'}, \xi(\sigma')$ have been defined so that

$B_{\sigma'} \subset A$ and $|B_{\sigma'}| = \beta_{\sigma'}$,

$\xi(\sigma')$ is a representative of $\{U_\xi : \xi \in B_{\sigma'}\}$, and

$\{U_{\xi(\sigma')} : \sigma' < \sigma\}$ has the finite intersection property.

We set

$C = \{\xi(\sigma') : \sigma' < \sigma\}$, and

$D = \{\xi \in A : \{U_\xi : \xi \in C \cup \{\xi\}\} \text{ has the finite intersection property}\}.$

Since

$$|C| \leq |\sigma| < \text{cf}(\beta) < \kappa \leq \kappa^\kappa,$$

it follows from the characteristic property of the set A (as given in Lemma 5.22) that $|D| = \alpha$. We choose $B_\sigma \subset D$ so that $|B_\sigma| = \beta_\sigma$ and, using 5.1 (applied to B_σ in place of α), we choose a representative $\xi(\sigma)$ of $\{U_\xi : \xi \in B_\sigma\}$. We note that since

$$\xi(\sigma) \in B_\sigma \subset D,$$

the family $\{U_{\xi(\sigma')} : \sigma' \leq \sigma\}$ has the finite intersection property.

The definition of the sets $\{B_\sigma : \sigma < \text{cf}(\beta)\}$, $\{\xi(\sigma) : \sigma < \text{cf}(\beta)\}$ is complete. It is clear that conditions (i), (ii) and (iii) are satisfied.

We set $B = \bigcup_{\sigma < \text{cf}(\beta)} B_\sigma$ and we note from Lemma 5.4, with α , $\text{cf}(\alpha)$, α_σ , $\{U_\xi : \xi < \alpha\}$ and $\{U_\xi : \xi < \alpha_\sigma\}$ replaced by β , $\text{cf}(\beta)$, β_σ , $\{U_\xi : \xi \in B\}$ and $\{U_\xi : \xi \in B_\sigma\}$, respectively, that there is a set S (this is the set A in the statement of Lemma 5.4) such that $S \subset B$, $|S| = \beta$, and

$$\{U_\xi : \xi \in S\} \text{ has the finite intersection property.}$$

The proof is complete.

5.26 Corollary. Assume the generalized continuum hypothesis. Let α and κ be infinite cardinals and let X be a compact, Hausdorff space such that $S(X) = \kappa$. Then

- (a) if $\text{cf}(\alpha) < \kappa$ then X does not have calibre α ;
- (b) if $\text{cf}(\alpha) > \kappa$, and if either $\text{cf}(\alpha)$ is a limit cardinal or $\text{cf}(\alpha) = \beta^+$ with $\text{cf}(\beta) \geq \kappa$, then X has calibre α ;
- (c) if $\text{cf}(\alpha) > \kappa$ and $\text{cf}(\alpha) = \beta^+$ with $\text{cf}(\beta) < \kappa$, then X has calibre (β^+, β) (and hence X has calibre (α, β)).

Proof. (a) It is clear that X does not have calibre $\text{cf}(\alpha)$; the required statement follows from Theorem 2.3(a).

(b) It follows from the assumptions (using Theorem A.8(b) of the Appendix in case $\text{cf}(\alpha) = \beta^+$ with $\text{cf}(\beta) \geq \kappa$) that $\kappa \ll \text{cf}(\alpha)$. If α is regular then $\kappa \ll \alpha$ and if α is singular then α is a (strong) limit cardinal and from $\kappa < \alpha$ it follows again that $\kappa \ll \alpha$. That X has calibre α now follows from Theorem 5.6.

(c) It follows from Theorem 5.25 (with $\text{cf}(\alpha)$ replacing α) that X

has calibre $(\beta^+, \omega^{\aleph/\text{cf}(\alpha)})$. For $\gamma < \beta$ and $\lambda < \kappa$ we have

$$\gamma^\lambda \leq 2^\gamma \cdot 2^\lambda \leq \beta \cdot \kappa < \text{cf}(\alpha)$$

and hence $\aleph/\sqrt{\text{cf}(\alpha)} \geq \beta$; hence X has calibre (β^+, β) .

Examples on calibres

The example given below in 5.28 proves, together with 6.14 for $\kappa = \omega^+$, with Kunen's example (7.9 and the remark following), with the generalized examples of Gaifman and of Argyros (6.19, 6.23 and the Notes for Chapter 6), and with the Laver–Galvin example of 7.13, that Corollary 5.26 cannot be improved in any substantive way; see the diagrams of 7.19. The example of 5.28 will be studied further in 6.29 below; a modification, providing answers to questions related to strictly positive measures, appears in 6.25 and 6.26.

5.27 Lemma. Let $\alpha \geq \omega$ and let X be a set with topologies $\mathcal{T}_1$ and $\mathcal{T}_2$ such that there is a base for $\mathcal{T}_1$ consisting of $\mathcal{T}_2$ -compact sets and such that

$$|X| = d(X, \mathcal{T}_1) = \alpha.$$

Then $\langle X, \mathcal{T}_1 \rangle$ does not have precalibre α .

Proof. Let $\langle x_\xi : \xi < \alpha \rangle$ be a well-ordering of the elements of X and for $\xi < \alpha$ let V_ξ be a $\mathcal{T}_2$ -compact, $\mathcal{T}_1$ -open subset of X such that

$$V_\xi \cap \{x_\zeta : \zeta \leq \xi\} = \emptyset.$$

If $\langle X, \mathcal{T}_1 \rangle$ has precalibre α there is $A \subset \alpha$ with $|A| = \alpha$ such that $\{V_\xi : \xi \in A\}$ has the finite intersection property, and since V_ξ is $\mathcal{T}_2$ -compact for $\xi \in A$ there is

$$x = x_\xi \in \bigcap_{\xi \in A} V_\xi.$$

There is $\xi \in A$ with $\bar{\xi} \leq \xi < \alpha$ and we have $x \notin V_\xi$, a contradiction.

Definitions of several terms used in the following theorem (tree, height, etc.) are given in Appendix A.

5.28 Theorem (Argyros). Assume the generalized continuum hypothesis. Let β be an (infinite) singular cardinal. There is a com-

pletely regular, Hausdorff space X such that

$$S(X) = (\text{cf}(\beta))^+, \quad \text{and} \\ X \text{ does not have precalibre } \beta^+.$$

Proof. There is a set $\{\beta_\sigma : \sigma < \text{cf}(\beta)\}$ such that

$$\begin{aligned} \beta_\sigma &\text{ is a regular cardinal for } \sigma < \text{cf}(\beta), \\ \omega \leq \beta_\sigma < \beta_\sigma^{++} \leq \beta_{\sigma+1} < \beta &\text{ for } \sigma < \text{cf}(\beta), \\ \prod_{\sigma < \tau} \beta_\sigma \leq \beta_\tau &\text{ for limit ordinals } \tau \text{ with } 0 < \tau < \text{cf}(\beta), \text{ and} \\ \sum_{\sigma < \text{cf}(\beta)} \beta_\sigma &= \beta. \end{aligned}$$

Let $\{B_\sigma : \sigma < \text{cf}(\beta)\}$ be a family of subsets of β such that

$$\begin{aligned} |B_\sigma| &= \beta_\sigma \quad \text{for } \sigma < \text{cf}(\beta), \\ B_\sigma \cap \beta_{\sigma'} &= \emptyset \quad \text{for } \sigma < \sigma' < \text{cf}(\beta), \quad \text{and} \\ \bigcup_{\sigma < \text{cf}(\beta)} B_\sigma &= \beta, \end{aligned}$$

and for $\sigma < \text{cf}(\beta)$ let $\{B_{\sigma+1,i} : i < \beta_\sigma\}$ be a family of subsets of $B_{\sigma+1}$ such that

$$\begin{aligned} |B_{\sigma+1,i}| &= \beta_{\sigma+1}, \quad \text{and} \\ B_{\sigma+1,i} \cap B_{\sigma+1,i'} &= \emptyset \quad \text{for } i < i' < \beta_\sigma. \end{aligned}$$

We define a tree $\langle T, \leq \rangle$ of height $\text{cf}(\beta)$ with

$$T = \bigcup_{\sigma < \text{cf}(\beta)} T_\sigma \subset [\beta]^2$$

We set $T_0 = [\beta_0]^2$.

Assume inductively that T_σ has been defined, with $T_\sigma \subset [B_\sigma]^2$ and $|T_\sigma| = \beta_\sigma$, and let $T_\sigma = \{s_i : i < \beta_\sigma\}$ be a one-to-one well-ordering of T_σ . We set

$$T_{\sigma+1} = \bigcup \{[B_{\sigma+1,i}]^2 : i < \beta_\sigma\}$$

and for $s = s_i \in T_\sigma$ and $t \in T_{\sigma+1}$ we define $s < t$ if and only if $t \in [B_{\sigma+1,i}]^2$.

Let τ be a limit ordinal with $0 < \tau < \text{cf}(\beta)$ and assume that $\{T_\sigma : \sigma < \tau\}$ have been defined, together with the ordering $\leq$ on $\bigcup_{\sigma < \tau} T_\sigma$, so that $\langle \bigcup_{\sigma < \tau} T_\sigma, \leq \rangle$ is a tree such that

$$T_\sigma = \{s : \{t : t < s\} \text{ has order-type } \sigma\}.$$

Let Σ_τ be the set of branches of length τ of this tree, and set $\beta_\tau^* = |\Sigma_\tau|$.

It is clear that

$$\beta_\tau^* \leq \prod_{\sigma < \tau} \beta_\sigma \leq \beta_\tau.$$

Let $\{\Sigma_i : i < \beta_\tau^*\}$ be a one-to-one well-ordering of Σ_τ , and let $\{B_{\tau,i} : i < \beta_\tau^*\}$ be a family of subsets of B_τ such that

$$\begin{aligned} |B_{\tau,i}| &= \beta_\tau \quad \text{for } i < \beta_\tau^*, \quad \text{and} \\ B_{\tau,i} \cap B_{\tau,i'} &= \emptyset \quad \text{for } i < i' < \beta_\tau^*. \end{aligned}$$

We set

$$T_\tau = \cup \{[B_{\tau,i}]^2 : i < \beta_\tau^*\},$$

and for $s \in \cup_{\sigma < \tau} T_\sigma$ and $t \in T_\tau$ we define $s < t$ if and only if there is $i < \beta_\tau^*$ such that $s \in \Sigma_i$ and $t \in [B_{\tau,i}]^2$.

The definition of the tree $\langle T, \leq \rangle$ is complete. We denote by Σ the set of all branches of T .

For $\sigma < \text{cf}(\beta)$ we set

$$\begin{aligned} A(\sigma) &= \cup \{s \in T : s \in \cup_{\sigma \leq \sigma' < \text{cf}(\beta)} T_{\sigma'}\}, \quad \text{and} \\ B(\sigma) &= \beta \setminus A(\sigma). \end{aligned}$$

For $s \in T$ we denote by K_s the set (of cardinality 2) of non-constant functions in $\{0, 1\}^s$; that is, if $s = \{\eta, \zeta\}$ then

$$K_s = \{\langle \eta, 1 \rangle, \langle \zeta, 0 \rangle\}, \{\langle \eta, 0 \rangle, \langle \zeta, 1 \rangle\}.$$

For $\Sigma \in \Sigma$ we set

$$\begin{aligned} A(\Sigma) &= \cup \{s : s \in \Sigma\}, \quad \text{and} \\ V_\Sigma &= \left(\prod_{s \in \Sigma} K_s \right) \times \{0, 1\}^{\beta \setminus A(\Sigma)} \end{aligned}$$

We now define the space $\langle X, \mathcal{T} \rangle$. As set, X is the set $\{0, 1\}^\beta$; the topology $\mathcal{T}$ is defined by the subbase that contains

- (i) all subsets of $\{0, 1\}^\beta$ of the form $\pi_j^{-1}(S) \times \{0, 1\}^{\beta \setminus J}$ with $|J| < \text{cf}(\beta)$ and with S compact in the usual product topology of $\{0, 1\}^J$; and
- (ii) all sets of the form V_Σ for $\Sigma \in \Sigma$.

The base $\mathcal{B}$ determined by this subbase consists of all sets of the form $U \cap (\cap_{j=1}^m V_{\Sigma_j})$ with $U = \pi_J^{-1}(S) \times \{0, 1\}^{\beta \setminus J}$ as above and with $\{\Sigma_j : 1 \leq j \leq m\} \subset \Sigma$. The elements of $\mathcal{B}$ are closed in the usual topology of $\{0, 1\}^\beta$, hence in its extension $\mathcal{T}$; thus $\langle X, \mathcal{T} \rangle$ is a completely regular, Hausdorff space.

We say that the basic set $V = U \cap (\cap_{j=1}^m V_{\Sigma_j})$ is *determined by* $(U; \Sigma_1, \dots, \Sigma_m)$, and we say that V is *separated at level σ* if

- (i) $(\Sigma_{j_1} \cup T_\sigma) \cap (\Sigma_{j_2} \cup T_\sigma) = \emptyset$ for $1 \leq j_1 < j_2 \leq m$, and
- (ii) there is $J \subset \beta$ with $U = \pi_J^{-1}(\pi_J[U])$ such that $J \cap A(\sigma) = \emptyset$.

If $V = U \cap (\cap_{j=1}^m V_{\Sigma_j}) \in \mathcal{B}$ and V is separated at level σ we set

$$V(0, \sigma) = \pi_{B(\sigma)}[V] \quad \text{and}$$

$$V(1, \sigma) = \bigcap_{j=1}^m \left(\left(\prod_{s \in \Sigma_j, s \subset A(\sigma)} K_s \right) \times \{0, 1\}^{A(\sigma) \setminus A(\Sigma_j)} \right).$$

We note that $V = V(0, \sigma) \times V(1, \sigma)$. We note also for $|J| < \text{cf}(\alpha)$ that $\{0, 1\}^J$ is discrete in the $\text{cf}(\alpha)$ -box topology; thus $V(0, \sigma)$ is open-and-closed in the $\text{cf}(\alpha)$ -box topology of $(\{0, 1\}^{B(\alpha)})_{\text{cf}(\alpha)}$.

For $\sigma < \text{cf}(\beta)$ we set

$$U(\sigma) = \{x \in X : x_\eta = 0 \text{ for } \eta < \sigma, x_\sigma = 1\}.$$

Then $\{U(\sigma) : \sigma < \text{cf}(\beta)\}$ is a faithfully indexed cellular family in the $\text{cf}(\beta)$ -box topology of $\{0, 1\}^\beta$, hence in the (larger) topology $\mathcal{T}$ of X ; thus $S(X) > \text{cf}(\beta)$.

To complete the proof that $S(X) = (\text{cf}(\beta))^+$ we show that in fact X has property $K_{(\text{cf}(\beta))^+}$. Let $\{V_\eta : \eta < (\text{cf}(\beta))^+\} \subset \mathcal{B}$; we assume without loss of generality that there is a (fixed) $\sigma < \text{cf}(\beta)$ such that each of the sets V_η is separated at level σ . For $J \subset B(\sigma)$ with $|J| < \text{cf}(\beta)$ we have from the generalized continuum hypothesis that $|\{0, 1\}^J| \leq \text{cf}(\beta)$ and hence $\{0, 1\}^J$ has calibre $(\text{cf}(\beta))^+$. Further from Theorem A.8(c) of Appendix A (with α replaced by $\text{cf}(\beta)$) we have from the generalized continuum hypothesis that $\text{cf}(\beta) \ll (\text{cf}(\beta))^+$. It then follows from Theorem 3.6(a), with α, β and γ replaced by $(\text{cf}(\beta))^+$ and with κ, I, X_i and Y replaced by $\text{cf}(\beta), B(\sigma), \{0, 1\}$ and $\{0, 1\}^{B(\sigma)}$ respectively, that $\{0, 1\}^{B(\sigma)}$ has calibre $(\text{cf}(\beta))^+$. Thus there is $A \subset (\text{cf}(\beta))^+$ with $|A| = (\text{cf}(\beta))^+$ such that

$$\bigcap_{\eta \in A} V_\eta(0, \sigma) \neq \emptyset.$$

If $\eta_1, \eta_2 \in A$ with $\eta_1 \neq \eta_2$ then every branch of T that determines V_{η_1} intersects at most one of the branches of T that determines V_{η_2} at level greater than or equal to σ ; hence

$$V_{\eta_1}(1, \sigma) \cap V_{\eta_2}(1, \sigma) \neq \emptyset.$$

It follows that every two elements of the family $\{V_\eta : \eta \in A\}$ have non-empty intersection, as required.

It remains to show that X does not have precalibre β^+ . Since the elements of the base $\mathcal{B}$ of X are compact in the usual product topology of $\{0, 1\}^\beta$, and since

$$|X| = 2^\beta = \beta^+$$

it is enough, according to Lemma 5.27, to show that $d(X) = \beta^+$.

Let $\{x_\xi : \xi < \beta\} \subset X$. We define a branch

$$\Sigma = \{s(\sigma) : \sigma < \text{cf}(\beta)\} \in \Sigma$$

such that if $\xi \in B_\sigma \subset \beta$ then $x_\xi \notin K_{s(\sigma)} \times \{0, 1\}^{\beta \setminus s(\sigma+1)}$.

We let $s(0) \in T_0$.

If $\sigma < \text{cf}(\beta)$ and $s(\sigma)$ has been defined there is $i < \beta_\sigma$ such that $s(\sigma) = s_i$ in the ordering of T_σ . For $\zeta \in B_{\sigma+1,i}$ we set

$$A_\zeta = \{\xi \in B_\sigma : x_\xi(\zeta) = 1\}$$

and we note that since $A_\zeta \in \mathcal{P}(B_\sigma)$ and

$$|\mathcal{P}(B_\sigma)| = 2^{\beta_\sigma} = \beta_\sigma^+ < \beta_{\sigma+1} = |B_{\sigma+1,i}|$$

there are distinct elements $\zeta_0, \zeta_1 \in B_{\sigma+1,i}$ such that $A_{\zeta_0} = A_{\zeta_1}$. We note that

$$x_\xi(\zeta_0) = x_\xi(\zeta_1) \quad \text{for all } \xi \in B_\sigma,$$

and we set $s(\sigma+1) = \{\zeta_0, \zeta_1\}$.

For non-zero limit ordinals $\tau < \text{cf}(\beta)$ we choose for $s(\tau)$ any element t of T_τ such that

$$s(\sigma) < t \quad \text{for all } \sigma < \tau.$$

The definition of the branch $\Sigma = \{s(\sigma) : \sigma < \text{cf}(\beta)\}$ is complete. It is clear that

$$B_\sigma \cap K_{s(\sigma)} \times \{0, 1\}^{\beta \setminus s(\sigma+1)} = \emptyset,$$

and hence that

$$V_\Sigma \cap \{x_\xi : \xi < \beta\} = \emptyset.$$

It follows that $\{x_\xi : \xi < \beta\}$ is not dense in X .

The proof is complete.

5.29 Corollary. Assume the generalized continuum hypothesis. Let κ and β be cardinals with κ regular and with $\text{cf}(\beta) < \kappa < \beta$. Then

there is an extremally disconnected, compact Hausdorff space Ω such that

$$S(\Omega) = \kappa, \quad \text{and} \\ \Omega \text{ does not have calibre } \beta^+.$$

Proof. From Theorem 5.28 there is a space X_0 such that $S(X_0) = (\text{cf}(\beta))^+$ and X_0 does not have precalibre β^+ , and from Theorem 3.30 there is a (completely regular, Hausdorff) space X_1 such that $S(X_1) = \kappa$. Clearly the disjoint union $X = X_0 \cup X_1$ does not have precalibre β^+ and satisfies $S(X) = \kappa$; we set $\Omega = G(X)$ (the Gleason space of X) and we note from Corollaries 2.18 and 2.22 that Ω is as required.

5.30 Problem (Argyros). Can one define in ZFC (without the continuum hypothesis or other special set-theoretic assumptions) a compact c.c.c. space that does not have calibre 2^ω ?

In 6.18 below we note (an improvement on) an observation of Erdős: Assuming $\omega^+ = 2^\omega$ there is a compact Hausdorff space with a strictly positive measure (hence, a c.c.c. space) that does not have calibre ω^+ .

Notes for Chapter 5

The material of 5.1–5.3, and Theorem 5.6 for regular cardinals α , is due to Argyros (1977a, b); 5.4 and 5.5, together with Theorem 5.6 for singular cardinals α , are due to Tsarpalias (1978a, b). Corollaries 5.8–5.10, which are consequences of Theorem 5.6, are from Argyros & Tsarpalias (1978a, b, 1981).

The results in the section concerning P_{λ^+} -spaces and λ^+ -box products (5.12–5.21) were noted by Negrepontis. Corollary 5.13 strengthens a result of Juhász (1971, Theorem 2.11), while Corollary 5.18 strengthens a result of Kurepa (1962) and Hedrlín (1966).

For $\kappa \geq \omega$ and X a space we set

$$\mathcal{G}_\kappa(X) = \{\cap \mathcal{U} : \mathcal{U} \subset \mathcal{T}(X), |\mathcal{U}| < \kappa\}.$$

We note a consequence of Theorem 5.12.

Theorem. Let $\omega \leq \lambda < \kappa \ll \alpha$ with α and κ regular, let X be a compact Hausdorff space such that $S(X) \leq \kappa$, let $\mathcal{F}$ be a set of con-

tinuous functions from X to a space Y such that $|Y| < \alpha$ and $\{p\} \in \mathcal{G}_{\lambda^+}(Y)$ for all $p \in Y$, and let $\Phi: \mathcal{F} \rightarrow Y$ with $\Phi(f) \in f[X]$ for $f \in \mathcal{F}$. If $|\mathcal{F}| \geq \alpha$ then there are $\mathcal{F}' \subset \mathcal{F}$ and $p \in Y$ such that

$$\begin{aligned} |\mathcal{F}'| &= \alpha, \\ \Phi(f) &= p \quad \text{for } f \in \mathcal{F}', \text{ and} \\ \cap \{f^{-1}(\{p\}) : f \in \mathcal{F}'\} &\neq \emptyset. \end{aligned}$$

Proof. Since $|Y| < \alpha$ and α is regular, we assume without loss of generality that there is $p \in Y$ such that

$$\Phi(f) = p \quad \text{for } f \in \mathcal{F}.$$

Since $f^{-1}(\{p\}) \in \mathcal{G}_{\lambda^+}(X) \subset \mathcal{T}(X_{(\lambda^+)})$ for $f \in \mathcal{F}$ and $X_{(\lambda^+)}$ has calibre α (by Theorem 5.12), there is $\mathcal{F}'$ as required.

Taking $\lambda = \omega$, $\kappa = \omega^+$, $\alpha = (2^\omega)^+$, $Y = \mathbb{R}$ and $\Phi = \sup$ we have the following statement: If G is a compact topological group and $\mathcal{F}$ a set of continuous functions from G to $\mathbb{R}$ with $|\mathcal{F}| > 2^\omega$, then there are $\mathcal{F}' \subset \mathcal{F}$, $\bar{x} \in G$ and $p \in \mathbb{R}$ such that

$$\begin{aligned} |\mathcal{F}'| &= (2^\omega)^+ \quad \text{and} \\ f(\bar{x}) &= \sup \{f(x) : x \in G\} = p \quad \text{for } f \in \mathcal{F}'. \end{aligned}$$

(That $S(G) = \omega^+$ follows from the fact that the Haar measure μ of G satisfies $\mu(G) = 1$ and $\mu(U) > 0$ for every non-empty, open subset U of G .) This property of compact groups (with $|\mathcal{F}'| = 2$) was observed several years ago by Juhász.

Lemma 5.22, Theorem 5.25 and Corollary 5.26 are due to Argyros & Tsarpalias (1978b, 1981)

Lemma 5.23 is noted by Rosenthal (1970a); the proof given here is due to Tsarpalias.

The results from functional analysis concerning the isomorphic embeddability of l_α^1 into the Banach spaces $C(X)$, discussed briefly in the introduction to this section, are given in Argyros (1977a, b), Tsarpalias (1978a, b), and Argyros & Tsarpalias (1978b). Analogous but more delicate results in functional analysis are available, using (in place of compact Hausdorff spaces X) P_{λ^+} -spaces and λ^+ -box products (as in 5.12–5.21 above).

Theorem 5.28, which establishes (assuming the generalized continuum hypothesis) a conjecture of Shelah (1980), is due to Argyros (1981c).

On the existence of independent families

The methods used in this chapter to determine calibres of compact spaces are closely connected with methods used to treat the following question. Let X be a totally disconnected, compact Hausdorff space of weight α ; under what conditions is there a continuous function from X onto $\{0, 1\}^\alpha$ (or, less strictly, onto $\{0, 1\}^\beta$ for some cardinal $\beta \leq \alpha$ as close to α as possible)?

A family $\{\langle A_\xi, B_\xi \rangle : \xi < \beta\}$ of ordered pairs of subsets of X , with $A_\xi \cap B_\xi = \emptyset$ for $\xi < \beta$, is called an *independent family* (of length β) if for every finite subset F of ω and every function $\varepsilon: F \rightarrow \{-1, +1\}$, we have

$$\bigcap_{\xi \in F} \varepsilon_\xi A_\xi \neq \emptyset$$

(where $(+1)A_\xi = A_\xi$, $(-1)A_\xi = B_\xi$ for $\xi < \beta$).

It is clear that the existence of a continuous function from X onto $\{0, 1\}^\beta$ is equivalent to the existence of an independent family $\{\langle A_\xi, B_\xi \rangle : \xi < \beta\}$ of length β such that A_ξ, B_ξ are closed in X for $\xi < \beta$.

We describe without proofs the current status of questions concerning the existence of continuous functions from X onto $\{0, 1\}^{wX}$. The conditions which arise are, in general, remarkably similar to those encountered in the study of calibres of compact spaces. There is, however, a striking exception: while the occurrence of the universal property of calibre has nothing to do with the completeness of the space (in the sense that a compact space X has calibre α if and only if its Gleason space $G(X)$ has calibre α), the occurrence of the (non-universal) property of the existence of a continuous function from X onto $\{0, 1\}^\alpha$ is influenced in a positive sense by the degree of completeness of X (cf. the theorem of Balcar & Franěk mentioned below in these Notes).

Theorem. Let α and κ be regular uncountable cardinals. If there is a compact, Hausdorff space Y such that $S(Y) = \kappa$ and $w(Y) \geq \alpha$, and if α is not a calibre of Y , then there is a space X such that

$$S(X) = \kappa \text{ and } w(X) = \alpha, \text{ and}$$

there is no continuous function from X onto $\{0, 1\}^\alpha$.

(This result is due to Argyros (1981b, Theorem 1.9).)

Corollary. Assume the generalized continuum hypothesis, and let α and κ be regular uncountable cardinals.

(a) If there is β such that $\alpha = \kappa = \beta^+$, then there is a totally disconnected, compact Hausdorff space X such that

$$S(X) = w(X) = \kappa, \quad \text{and}$$

there is no continuous function from X onto $\{0, 1\}^\kappa$.

(b) Assume in addition that $V = L$, that $\alpha = \kappa$, and that the cardinal α is strongly inaccessible but not weakly compact. Then there is a totally disconnected, compact Hausdorff space X such that

$$S(X) = w(X) = \kappa, \quad \text{and}$$

there is no continuous function from X onto $\{0, 1\}^\kappa$.

(c) If there is a cardinal number β such that $\text{cf}(\beta) < \kappa < \beta < \beta^+ = \alpha$, then there is a totally disconnected, compact Hausdorff space X such that

$$S(X) = \kappa \quad \text{and} \quad w(X) = \alpha, \quad \text{and}$$

there is no continuous function from X onto $\{0, 1\}^\alpha$.

(For (a) we use the generalized Gaiĭman example (6.23, Notes for Chapter 6), or the generalized Argyros example (6.26, Notes for Chapter 6), or the example of Kunen (7.9), or the example of Laver & Galvin (7.13), or for $\kappa = \omega^+$ the example of Erdős (6.18). For (b) we use the example of Jensen (Notes for Chapter 7). For (c) we use Argyros' example (5.28), or for $\kappa = \omega^+$ the example of Argyros & Tsarpalias (6.18).

Theorem. Let α and κ be regular cardinals with $\kappa \leq \alpha$, and let X be a totally disconnected, compact Hausdorff space such that $S(X) = \kappa$. Then there is a continuous function from X onto $\{0, 1\}^\alpha$.

(This result has been proved by Argyros & Tsarpalias (1978a).)

Corollary. Assume the generalized continuum hypothesis. If either $\kappa \leq \text{cf}(\beta) \leq \beta < \beta^+ = \alpha$ or α is a regular limit cardinal such that $\alpha > \kappa$, and if X is a totally disconnected, compact Hausdorff space with $S(X) = \kappa$ and $w(X) = \alpha$, then there is a continuous function from X onto $\{0, 1\}^\alpha$.

Theorem. If κ is a weakly compact cardinal and X is a totally

disconnected, compact Hausdorff space such that $S(X) = w(X) = \kappa$, then there is a continuous function from X onto $\{0, 1\}^\kappa$.

(This result follows directly from the methods of Argyros & Tsarpalias (1978a, 1981); it has also been noted by Shelah (1980).)

We turn now to results concerning singular cardinals.

Theorem. Assume the generalized continuum hypothesis, and let α be an (infinite) singular cardinal. If X is a totally disconnected, compact Hausdorff space such that

$$w(X) \geq \alpha \quad \text{and} \quad \text{cf}(\alpha) \text{ is a calibre of } X,$$

then there is a continuous function from X onto $\{0, 1\}^\alpha$.

(This result is due to Tsarpalias (1978); cf. also Argyros & Tsarpalias (1978a, 1981).)

Theorem. Let α be an (infinite) singular cardinal. If there is a totally disconnected, compact Hausdorff space Y such that $S(Y) = \kappa$ and $w(Y) \geq \alpha$, and if $\text{cf}(\alpha)$ is not a calibre of Y , then there is a totally disconnected, compact Hausdorff space X such that

$$S(X) = \kappa \quad \text{and} \quad w(X) = \alpha, \text{ and}$$

there is no continuous function from X onto $\{0, 1\}^\alpha$.

(This result, as well as several of the theorems of Argyros & Tsarpalias (concerning precalibres, calibres of compact spaces, and independent families) described in Chapter 5 and above in these Notes, has been obtained by Shelah (1980, Section 4.7).)

Theorem. Let α be an infinite cardinal. If X is an extremally disconnected, compact Hausdorff space with $w(X) = \alpha$, then there is a continuous function from X onto $\{0, 1\}^\alpha$.

(This result is due to Balcar & Franěk (1981).)

Definition. Let X be a space and $x \in X$. The *pseudoweight* of X at x , denoted $\pi(X, x)$, is the least cardinal number α for which there is a set $\mathcal{B}$ of non-empty, open subsets of X such that

$$|\mathcal{B}| = \alpha, \quad \text{and}$$

if U is open in X and $x \in U$ then there is $B \in \mathcal{B}$ such that $B \subset U$.

Theorem. Let α be an infinite cardinal and X a totally disconnected, compact Hausdorff space. There is a continuous function from X onto $\{0, 1\}^\alpha$ if and only if there is a non-empty, closed subset F of X such that $\pi(F, x) \geq \alpha$ for all $x \in F$.

(This result is due to Šapírovskii (1975); see also Juhász (1980, Section 3.18) for a complete account.)

The results described above provide, at least assuming the generalized continuum hypothesis, a nearly complete answer to the question of the existence of continuous functions from totally disconnected, compact Hausdorff spaces onto spaces of the form $\{0, 1\}^\alpha$. There remains to be considered the question of the highest degree of completeness possible for a space X (as above) that does not map continuously onto $\{0, 1\}^{w(X)}$. More specifically, we formulate a (standard) definition and pose a question.

Definition. Let β be an infinite cardinal and X a space with a base of open-and-closed subsets. The space X is a β -*extremally disconnected* if for every set U that is the union of fewer than β open-and-closed subsets of X , the set $\text{cl}_X U$ is open in X .

Problem. Let α and κ be regular, uncountable cardinals for which there is a totally disconnected, compact Hausdorff space X such that $S(X) = \kappa$, $w(X) \geq \alpha$, and α is not a calibre of X . What is the least cardinal β for which there is no β -extremally disconnected, compact Hausdorff space X such that

$$S(X) = \kappa \quad \text{and} \quad w(X) = \alpha, \text{ and}$$

there is no continuous function from X onto $\{0, 1\}^\alpha$?

(We note that it follows from the theorem of Balcar & Franěk that $\beta \leq \alpha$.)

The various results and questions above are easily modified to yield analogues concerning the existence of continuous functions from compact Hausdorff spaces onto spaces of the form $[0, 1]^\alpha$.

On the isomorphic theory of Banach spaces

Some of the results cited above have significant applications to certain central questions in the isomorphic theory of Banach spaces.

We now describe some of these applications. We note that most of the results concerning the isomorphic embeddability of the spaces $l^1(\Gamma)$ into Banach spaces of the form $C(X)$ depend on a criterion of Rosenthal (1974b) involving the existence in X of an independent family of length $|\Gamma|$.

Theorem. Let α and κ be (infinite) regular cardinals with $\kappa \ll \alpha$ and let X be a compact, Hausdorff space such that $S(X) \leq \kappa$. If B is a closed linear subspace of $C(X)$ such that $d(B) \geq \alpha$, then there is an isomorphic embedding of $l^1(\Gamma)$ into B .

(This follows from the methods of Argyros (1977a, b), Tsarpalias (1978a, b), and Argyros & Tsarpalias (1978b). Analogous but more delicate results, using (in place of compact Hausdorff spaces X) P_{λ^+} -spaces and λ^+ -box products as in 5.12–5.21 above, have been given by N. Kalamidas (1981), Th. Zachariades (1981), and Negreponitis (1980).)

The theorem just cited, together with the results of Rosenthal (1970a, p. 230) connecting the Souslin number of a compact space X with the density character of the weakly compact subsets of the Banach space $C(X)$, suggests the following question; here for a Banach space B we denote by $\Sigma(B)$ the cardinal number

$$\Sigma(B) = \min\{\lambda: \text{no weakly compact } K \subset B \text{ satisfies } d(K) \geq \lambda\}.$$

Problem (Argyros & Negreponitis). Let B be a Banach space. Set $\alpha = \dim B$ and $\kappa = \Sigma(B)$ and suppose $\kappa \ll \alpha$ and $\kappa \ll \text{cf}(\alpha)$. Does it follow that there is an embedding of $l^1(\alpha)$ into B ?

It is clear from the foregoing results that for a totally disconnected, compact Hausdorff space X , these three properties are closely related: X has calibre α ; there is a continuous function from X onto $\{0, 1\}^\alpha$; and there is an isomorphic embedding of $l^1(\alpha)$ into every closed, linear subspace B of $C(X)$ such that $d(B) \geq \alpha$. The following theorem limits our attempts to prove general theorems stronger than those mentioned above.

Theorem. Assume the continuum hypothesis. There is a totally disconnected, compact Hausdorff c.c.c. space X such that X does not have calibre ω^+ , there is no continuous function from X onto

$\{0, 1\}^{\omega^+}$, and there is an isomorphic embedding of $l^1(\omega^+)$ into $C(X)$ – indeed for every closed, linear non-separable subspace B of $C(X)$ such that B is generated by characteristic functions of open-and-closed subsets of X , there is an isomorphic embedding of $l^1(\omega^+)$ into B .

(This example, due to Argyros & Kalamidas, appears in Kalamidas (Doctoral dissertation, 1981); a similar result has been obtained by R. Haydon (unpublished).)

A Banach space B is *injective* if for every Banach space E and every closed, linear subspace F of E , every bounded linear transformation $T:F \rightarrow B$ extends to a bounded linear transformation $S:E \rightarrow B$. If in addition it is required that $\|S\| = \|T\|$, then B is called a $\mathcal{P}_1$ -space.

It is known that a Banach space B is a $\mathcal{P}_1$ -space if and only if B is isomorphic to $C(X)$ for some extremally disconnected, compact Hausdorff space X (cf. Goodner 1950, Kelley 1954, Nachbin 1950).

No characterization of injective Banach spaces is known, and in fact there is no known example of an injective Banach space that is not a $\mathcal{P}_1$ -space; Rosenthal (1970b) has remarked, however, that if there is an Ulam-measurable cardinal then there is a compact F -space X such that $C(X)$ is an injective Banach space and is not a $\mathcal{P}_1$ -space.

It is conjectured by Rosenthal (1970a) that if B is an injective Banach space with $d(B) = \alpha$, then there is an isomorphic embedding of $l^1(\alpha)$ into B . The theorem of Balcar and Franěk mentioned above settles this conjecture affirmatively for $\mathcal{P}_1$ -spaces B ; additional interesting partial results are given by Argyros (1981a).

A more detailed survey of some of the applications of independent families to the isomorphic theory of Banach spaces is given by Negreponitis (1980). Other recent surveys in the isomorphic theory of Banach spaces, dealing in part with techniques involving infinitary combinatorics, are given by Rosenthal (1975, 1978), Pełczyński (1979), and Haydon (1980).